Malthus Meets Romer: Population Growth Under Information Frictions

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Abstract

Households disagree about whether larger populations enhance productivity (Romerians) or diminish it (Malthusians). I embed this disagreement into a Barro-Becker fertility model with global games. Wrong beliefs generate different pathologies: in Romerian worlds, Malthusians cause prolonged stagnation through an *informational poverty trap*; in Malthusian worlds, Romerians cause boom-bust cycles. Optimal policy depends critically on the welfare criterion; I characterize how conclusions differ under dynastic, paternalistic, and utilitarian objectives. Survey data confirm that Malthusian beliefs significantly correlate with lower fertility.

JEL Classification: D83, J13, O40, C72

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1 Introduction

How many children should a family have? The answer depends critically on beliefs about the economic consequences of population size. A parent who believes, in the spirit of Romer (1990), that more people generate more ideas and faster growth might view an additional child as contributing to future prosperity. A parent who fears, in the spirit of Malthus (1798), that more people strain finite resources might view the same child as hastening collective impoverishment. These beliefs directly shape fertility decisions under intergenerational altruism, and the resulting population dynamics feed back into productivity outcomes that either validate or refute the beliefs. This paper studies the consequences of this feedback loop when households hold heterogeneous beliefs about the population-productivity relationship.

To formalize this idea, I develop an overlapping generations model in which households make fertility decisions under uncertainty about a key structural parameter: the scale effect of population on productivity growth. Households observe the same aggregate data but interpret it through different priors. Some households are "Romerians" who believe that larger populations accelerate innovation and growth. Others are "Malthusians" who believe that larger populations depress living standards by diluting fixed factors. These beliefs are transmitted across generations through a combination of cultural inheritance and Bayesian updating from observed outcomes.

The model rests on a single behavioral premise, namely that beliefs about population's economic consequences shape fertility, and it is worth establishing this premise empirically before drawing out its theoretical implications. Using data from the General Social Survey (GSS), I document robust negative correlations between Malthusian beliefs, measured by agreement with the statement that Earth cannot continue to support population growth, and both ideal and actual fertility. Individuals who hold Malthusian beliefs report ideal fertility approximately 0.27 children lower than non-Malthusians and have 0.29 fewer children, differences of roughly 11 and 16 percent relative to the respective sample means. These relationships persist after controlling for age, education, sex, and survey year, and are present across demographic groups. Tellingly, the effect on *actual* fertility is at least as large as the effect on stated preferences, suggesting that beliefs translate into consequential behavior rather than cheap talk. With this empirical foundation in place, I turn to the theoretical analysis of how belief heterogeneity shapes long-run dynamics.

The central theoretical contribution is to show that wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. This asymmetry arises from a simple but underappreciated feature of Bayesian learning about scale effects: the informativeness of productivity signals depends on population size. When population is near a reference level (normalized to one), productivity growth reveals almost nothing about whether scale effects are positive or negative. Learning therefore requires population to deviate substantially from this uninformative region, but such deviations require the very optimistic beliefs that pessimism suppresses. This interaction between beliefs, fertility, and the informativeness of data is the engine driving the paper's main results.

Consider first a Romerian world where scale effects are truly positive. Here Malthusian minorities do not merely slow convergence to efficiency. They can trap the economy in a state of prolonged stagnation. Pessimistic beliefs suppress fertility, keeping population low. Low population generates uninformative signals about the population-productivity relationship. Uninformative signals fail to correct pessimistic beliefs. The trap is self-reinforcing: the economy remains stuck not because Malthusianism is correct, but because population never grows large enough to reveal that it is wrong. I call this the *informational poverty trap*. While eventual convergence to correct beliefs occurs almost surely, the expected escape time can exceed any relevant historical horizon.

Now consider the mirror case: a Malthusian world where scale effects are truly negative. Here the dynamics are starkly different, generating boom-bust cycles rather than traps. Demographic selection always favors optimists, who have more children regardless of whether their optimism is warranted. But learning favors pessimists when population growth leads to poor outcomes. These opposing forces generate endogenous oscillations: during expansions, optimists outbreed pessimists; population grows; eventually the economy hits constraints; outcomes deteriorate; learning corrects beliefs toward Malthusianism; fertility falls; recovery begins; and the cycle repeats.

Taken together, the two cases reveal a sharp duality: *in Romerian worlds, Malthusians cause stagnation; in Malthusian worlds, Romerians cause instability.* Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.

This duality is more than a theoretical curiosity. It provides a unified lens for interpreting demographic history, though one that admits two competing interpretations of the pre-industrial Malthusian era. The conventional interpretation is that the world was genuinely Malthusian before industrialization: scale effects were negative, and the cycles documented by economic historians (Clark, 2007; Voigtländer and Voth, 2013) reflect the boom-bust dynamics of a true Malthusian regime. Under this view, the Industrial Revolution marked a genuine technological regime change from negative to positive scale effects. The alternative interpretation, enabled by the informational poverty trap, is that the world was always Romerian, but humanity was stuck in a trap where population never grew large enough to generate the information needed to escape pessimism. On this view, the Industrial Revolution was not a technological regime change but an *informational escape*: some combination of circumstances pushed population into the informative region, beliefs updated toward Romerianism, and the trap dissolved. Both interpretations are consistent with the historical evidence, and distinguishing them has profound implications for understanding the modern fertility decline, which the model interprets variously as a partial drift back toward the trap, an efficient response to a renewed Malthusian regime, or a quantity-quality substitution unrelated to scale effects.

If wrong beliefs can trap economies in stagnation or generate damaging cycles, a natural question is whether policy can help, and what "help" even means when the population itself is endogenous to the policy. Addressing this question requires careful attention to welfare evaluation, since, as Golosov et al. (2007) emphasize, standard welfare criteria become problematic when the set of agents depends on policy. I analyze optimal policy under three welfare criteria: dynastic welfare that respects Barro-Becker preferences, paternalistic welfare that evaluates outcomes under the true model, and utilitarian welfare that aggregates across all individuals. The optimal announcement policy differs substantially across criteria. A dynastic planner favors truthful communication, a paternalistic planner shades announcements toward the believed truth, and a utilitarian planner most aggressively promotes high fertility in Romerian worlds. Normative conclusions about the costs of wrong beliefs thus depend critically on the welfare criterion employed.

Beyond the choice of welfare criterion, the analysis of public information itself reveals a novel asymmetry. In standard coordination games, public information can be harmful by inducing excessive coordination on potentially incorrect signals (Morris and Shin, 2002). In my setting, the welfare implications depend on where the economy sits. When the economy is trapped in the uninformative region, public information is most valuable, since a credible announcement can coordinate beliefs and push population into the informative region, initiating escape from the trap. But this is precisely when the planner faces maximum uncertainty, since the same uninformative signals that trap households also limit what the planner can learn. The planner's dilemma is sharpest when intervention matters most.

This tension yields a policy prescription that differs from standard results. In the informational poverty trap, even a moderately noisy announcement can improve welfare by breaking the self-reinforcing pessimism, whereas in a Malthusian world with cycles, announcements require higher precision to avoid amplifying volatility. The heterogeneous strategic interaction, with Romerians exhibiting strategic complementarity and Malthusians exhibiting strategic substitutability, provides a natural stabilizer that makes public information less dangerous than in pure-complementarity environments, though it does not eliminate the risk entirely.

Related Literature

This paper connects several strands of literature: endogenous fertility and population, heterogeneous beliefs in macroeconomics, global games, and cultural transmission.

The foundational work on endogenous fertility is Becker (1960), who introduced the quantity-quality tradeoff, and Barro and Becker (1989), who embedded fertility choice in a dynastic optimization framework. Subsequent contributions have enriched this framework with human capital accumulation (Galor and Weil, 2000), mortality transitions (Kalemli-Ozcan, 2003), and intergenerational transfers (Lee, 2003). The unified growth theory of Galor (2011) synthesizes these elements to explain the demographic transition from Malthusian stagnation to modern growth. My contribution differs by introducing belief heterogeneity as a driver of fertility differentials and population dynamics, rather than focusing on the evolution of technology or human capital.

The debate over population scale effects has a long history. Malthus (1798) articulated the view that population growth outstrips resource growth, leading to immiseration. Romer (1990) and Kremer (1993) formalized the opposing view that larger populations generate more ideas, accelerating technological progress. Jones (1995) and Jones (2022) provide frameworks that nest both perspectives. Empirical work by Ashraf and Galor (2013) finds evidence for positive scale effects over very long horizons, while Voigtländer and Voth (2013) documents Malthusian dynamics in pre-industrial Europe. I do not take a stand on which view is correct; rather, I study the consequences of households holding different views.

The literature on heterogeneous beliefs has grown substantially. In macroeconomics, Angeletos and La'O (2013) and Angeletos et al. (2018) examine how dispersed information affects business cycles and sentiments. Bordalo et al. (2018) develop diagnostic expectations with belief overreaction. Malmendier and Nagel (2011) document that personal experiences shape macroeconomic expectations. In finance, Simsek (2013), Geanakoplos (2010), and Scheinkman and Xiong (2003) study asset pricing under disagreement. My paper differs by modeling disagreement about structural parameters governing long-run relationships rather than beliefs about current states or short-run forecasts.

The global games literature, pioneered by Carlsson and Van Damme (1993) and developed by Morris and Shin (1998) and Morris and Shin (2003), provides tools for analyzing games with strategic complementarities under incomplete information. I adapt these techniques to a setting with both strategic complementarities (among Romerians) and strategic substitutabilities (among Malthusians), which creates novel analytical challenges but also generates richer dynamics than standard global games applications such as currency crises (Morris and Shin, 1998) or bank runs (Goldstein and Pauzner, 2005).

The literature on cultural transmission, developed by Cavalli-Sforza and Feldman (1981), Bisin and Verdier (2001), and Bisin and Verdier (2011), provides frameworks for understanding how preferences and beliefs persist across generations. Doepke and Zilibotti (2019) apply these tools to study parenting styles. I incorporate cultural transmission as one channel through which beliefs evolve, alongside Bayesian learning from observed outcomes.

The analysis of public information in coordination games relates to work by Morris and Shin (2002) on the social value of public information. They show that in games with strategic complementarities, public information can be harmful if it crowds out private information. In my setting, the welfare implications of public announcements are more nuanced because of the heterogeneous strategic interaction and the state-dependent value of coordination: public information is most valuable in the trap, where it is also most uncertain.

An important related literature addresses welfare evaluation with endogenous population. Golosov et al. (2007) provide a systematic analysis of efficiency concepts when fertility is a choice variable, showing that standard welfare theorems fail under total or average utilitarianism but that a modified Pareto criterion can restore efficiency of competitive equilibria. Blackorby et al. (2005) develop the axiomatic foundations of population ethics and characterize critical-level utilitarianism as a resolution to the "Repugnant Conclusion" of Parfit (1984). I build on these insights by analyzing how different welfare criteria affect optimal communication policy in a setting with belief heterogeneity.

Finally, my paper relates to work on poverty traps and multiple equilibria in development economics. Azariadis and Drazen (1990) formalize poverty traps arising from threshold externalities. Banerjee and Newman (1993) study occupational choice with wealth constraints. The informational poverty trap I identify differs from these mechanisms: the trap arises not from technological nonconvexities or credit constraints but from the endogenous uninformativeness of signals when the economy remains in a particular region of the state space.

The paper proceeds as follows. Section 2 presents the model environment, including the fertility choice problem, belief formation, and intergenerational transmission. Section 3 presents empirical evidence validating the model's central prediction that Malthusian beliefs reduce fertility. Section 4 analyzes strategic complementarity and substitutability using global games techniques. Section 5 examines long-run dynamics under learning and demographic selection, including the informational poverty trap. Section 6 discusses the alignment between theoretical predictions and demographic history, offering two competing interpretations. Section 7 analyzes the value of public information, characterizes optimal communication policy under alternative welfare criteria, and compares decentralized and planner allocations. Section 8 concludes. All proofs, along with auxiliary theoretical results, are collected in the Appendix.

2 Model

I develop a discrete-time overlapping generations model with dynastic altruism. The economy is populated by a continuum of households who make fertility decisions. Households differ in their beliefs about the structural relationship between population and productivity growth.

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \ldots$. Each period, a generation of adults makes economic decisions; their children become adults in the following period.

Production. A single final good is produced using labor according to the technology

$$Y_t = A_t N_t^{\alpha}, \tag{1}$$

where Y_t is output, A_t is total factor productivity (TFP), N_t is population (equivalently, labor supply), and $\alpha \in (0, 1)$ captures diminishing returns to labor given fixed factors such as land.

TFP evolves according to

$$A_{t+1} = A_t \left(1 + g(N_t; \theta)\right), \tag{2}$$

where the productivity growth function is

$$g(N_t; \theta) = \bar{g} + \theta \ln N_t. \tag{3}$$

Here $\bar{g} > 0$ is baseline productivity growth and $\theta \in \mathbb{R}$ is the *scale effect parameter*. When $\theta > 0$, larger populations generate faster productivity growth through mechanisms such as increased idea production (Romer, 1990; Kremer, 1993). When $\theta < 0$, larger populations slow productivity growth, perhaps through environmental degradation or resource depletion. When $\theta = 0$, productivity growth is independent of population.

The specification $g(N_t; \theta) = \bar{g} + \theta \ln N_t$ has an important implication for learning: when $N_t = 1$, productivity growth equals $\bar{g}$ regardless of θ. Observations near $N_t = 1$ are therefore *uninformative* about the scale effect parameter. This feature is central to the informational poverty trap developed in Section 5.

Per capita consumption, which I take as the welfare-relevant variable, is

$$c_t = \frac{Y_t}{N_t} = A_t N_t^{\alpha-1}. \tag{4}$$

Since $\alpha < 1$, larger populations reduce current per capita consumption holding TFP fixed. However, if $\theta > 0$, larger populations raise future TFP, creating an intertemporal tradeoff. The welfare implications of population growth depend on the relative magnitudes of these effects.

Households and Beliefs. The economy contains a unit mass of dynastic households indexed by $i \in [0, 1]$. Each household i holds a belief θ_i about the true value of the scale effect parameter θ. Beliefs are heterogeneous across households.

Assumption 1 (Belief Types)**.** *There exist two types of households. A fraction $\phi_t \in [0, 1]$ are* Romerians *with belief $\theta_R > 0$. The remaining fraction $1 - \phi_t$ are* Malthusians *with belief $\theta_M < 0$.*

I denote the true value of the scale effect parameter by $\theta^* \in \mathbb{R}$. A "Romerian world" has $\theta^* > 0$; a "Malthusian world" has $\theta^* < 0$. Households do not know θ^* with certainty but hold prior beliefs and update using Bayes' Rule.

The distribution of beliefs, characterized by ϕ_t, is common knowledge in each period, but individual household types are private information. Households observe aggregate variables (Y_t, A_t, N_t) and the history thereof, but do not observe other households' beliefs or fertility choices directly.

Fertility Choice. Adult households choose consumption c_{it} and fertility $n_{it} \geq 0$ (children per adult). The household faces the budget constraint

$$c_{it} + \psi(n_{it}) = w_t, \tag{5}$$

where $w_t = \alpha A_t N_t^{\alpha-1}$ is the wage (marginal product of labor) and $\psi : \mathbb{R}_+ \to \mathbb{R}_+$ is the cost of raising children.

Assumption 2 (Child-Rearing Cost)**.** *The child-rearing cost function satisfies $\psi(0) = 0$, $\psi'(n) > 0$, $\psi''(n) > 0$ for all $n \geq 0$, and $\lim_{n\to\infty} \psi'(n) = \infty$.*

For analytical tractability, I use the quadratic specification $\psi(n) = \frac{\gamma}{2} n^2$ with $\gamma > 0$ in most of the analysis.

Households have dynastic preferences. Following Barro and Becker (1989), I assume the household derives utility from own consumption and from the welfare of descendants:

$$U_{it} = u(c_{it}) + \beta n_{it} \mathbb{E}_{it} \left[V_{i,t+1} \right], \tag{6}$$

where $u(\cdot)$ is a strictly concave period utility function (I use $u(c) = \ln c$), $\beta \in (0,1)$ is the discount factor, and $V_{i,t+1}$ is the continuation value for the dynasty starting next period. The term n_{it} multiplying the continuation value captures the idea that the parent's utility from future generations scales with the number of children; this is the Barro-Becker formulation. Throughout, I assume that the equilibrium fertility levels satisfy $\beta n_j^* < 1$ so that the dynastic value function is well defined.

The expectation $\mathbb{E}_{it}[\cdot]$ is taken under household i's subjective beliefs about the parameter θ and about the actions of other households.

Population Dynamics. Aggregate population evolves according to

$$N_{t+1} = \bar{n}_t \cdot N_t, \tag{7}$$

where $\bar{n}_t$ is average fertility:

$$\bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}. \tag{8}$$

Here $n_{R,t}$ and $n_{M,t}$ denote the fertility choices of Romerians and Malthusians, respectively.

2.2 Belief Formation and Intergenerational Transmission

Beliefs evolve across generations through two channels: cultural transmission from parents and learning from observed economic outcomes.

Cultural Transmission. Children are born to households and initially adopt their parents' beliefs with probability $\lambda \in [0,1]$. With probability $1 - \lambda$, they form beliefs independently through learning.

Assumption 3 (Cultural Transmission)**.** *A child born to a type-j household adopts type j with probability λ and forms beliefs through learning with probability $1 - \lambda$.*

When $\lambda = 1$, beliefs are perfectly persistent across generations. When $\lambda = 0$, beliefs are formed entirely through learning.

Bayesian Learning. Children who do not inherit parental beliefs form beliefs by applying Bayes' rule to observed data. Let π_t denote the common prior probability that the world is Romerian (i.e., $\theta^* = \theta_R$) before observing generation-t outcomes. I assume that children who learn observe realized productivity growth:

$$s_t = g(N_t; \theta^*) + \varepsilon_t = \bar{g} + \theta^* \ln N_t + \varepsilon_t, \tag{9}$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is an i.i.d. noise term representing measurement error or transitory productivity shocks that obscure the true relationship.

Given signal s_t and population N_t, the likelihood of observing s_t under belief θ is

$$\ell(s_t|\theta) = \frac{1}{\sqrt{2\pi}\sigma_\varepsilon} \exp\left(-\frac{(s_t - \bar{g} - \theta \ln N_t)^2}{2\sigma_\varepsilon^2}\right). \tag{10}$$

Bayes' rule gives the posterior:

$$\pi_{t+1} = \frac{\pi_t \cdot \ell(s_t|\theta_R)}{\pi_t \cdot \ell(s_t|\theta_R) + (1 - \pi_t) \cdot \ell(s_t|\theta_M)}. \tag{11}$$

Children who learn adopt Romerian beliefs with probability π_{t+1} and Malthusian beliefs with probability $1 - \pi_{t+1}$.

Signal Informativeness. A key feature of the learning technology is that signal informativeness depends on population. The expected log-likelihood ratio under the true model θ^* is:

$$\mathcal{I}_t \equiv \mathbb{E}^{\theta^*}\left[\ln \frac{\ell(s_t|\theta^*)}{\ell(s_t|\theta')}\right] = \frac{(\theta^* - \theta')^2 (\ln N_t)^2}{2\sigma_\varepsilon^2}, \tag{12}$$

for any alternative belief $\theta' \neq \theta^*$. This is the Kullback-Leibler divergence between the signal distributions under the two beliefs. When $N_t = 1$ (so $\ln N_t = 0$), the KL divergence is zero: the signal is completely uninformative about θ^*. As $|\ln N_t|$ grows, the signal becomes increasingly informative.

This dependence of informativeness on population is the source of the informational poverty trap. If pessimistic beliefs keep population near 1, signals remain uninformative, and pessimism persists even if incorrect.

Evolution of Belief Distribution. Combining cultural transmission and learning, the Romerian share evolves as follows. The fraction of next-generation adults who are Romerian children of Romerian parents is $\phi_t^{birth} \cdot \lambda$, where

$$\phi_t^{birth} = \frac{\phi_t n_{R,t}}{\bar{n}_t} \tag{13}$$

is the fraction of children born to Romerians. The fraction who are learning children of any parent who learn Romerian is $(1 - \lambda) \cdot \pi_{t+1}$. Summing:

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda)\pi_{t+1}. \tag{14}$$

This law of motion shows that the Romerian share is pushed upward by differential fertility (through ϕ_t^{birth}, which exceeds ϕ_t when $n_R > n_M$) and by learning (through π_{t+1}, which moves toward 1 when evidence favors Romerians).

2.3 Equilibrium

I now define equilibrium in this economy.

Definition 1 (Equilibrium)**.** *An equilibrium consists of:*

(i) Fertility policy functions $n_R(A_t, N_t, \phi_t)$ *and* $n_M(A_t, N_t, \phi_t)$ *for each type;*

(ii) Aggregate fertility $\bar{n}(A_t, N_t, \phi_t) = \phi_t n_R + (1 - \phi_t) n_M$*;*

(iii) Laws of motion for $(A_{t+1}, N_{t+1}, \phi_{t+1})$ *given by* (2), (7), *and* (14)*;*

such that:

(a) Each household's fertility choice maximizes expected dynastic utility (6) *subject to* (5)*, given beliefs* θ_i *and given the aggregate fertility function* $\bar{n}(\cdot)$*;*

(b) Expectations about aggregate fertility are consistent with the equilibrium aggregate fertility function;

(c) The actual laws of motion use the true parameter θ^**.*

Two features of this equilibrium concept merit emphasis. First, households optimize under their subjective beliefs but the actual economy evolves according to the true parameter θ^*. This allows for systematic forecast errors. Second, households must form expectations about aggregate fertility, which depends on other households' beliefs and actions. I impose rational expectations about others' behavior: each household correctly understands the mapping from the state to aggregate fertility, even though households disagree about structural parameters.

I emphasize that this notion of equilibrium describes a *decentralized* allocation in which individual households take aggregate fertility as given and do not internalize the externality that population growth imposes on TFP through (2). Even when all households share the correct belief θ^*, the resulting allocation is therefore not in general the social planner's optimum—a point that becomes important when interpreting the welfare benchmark in Section 7.

2.4 Household Optimization

I now characterize the household's optimal fertility choice. Let $V_j(A, N, \phi)$ denote the value function for a type-j household. The Bellman equation is:

$$V_j(A, N, \phi) = \max_{n \geq 0} \left\{ \ln(w - \psi(n)) + \beta n \cdot \mathbb{E}_j \left[V_j(A', N', \phi') \right] \right\}, \tag{15}$$

where the expectation is taken under belief θ_j and where (A', N', ϕ') evolve according to the laws of motion.

I conjecture a value function of the form:

$$V_j(A, N, \phi) = v_j(\phi) + \xi_{A,j} \ln A + \xi_{N,j} \ln N, \tag{16}$$

where $\xi_{A,j}$ and $\xi_{N,j}$ are type-specific constants (because beliefs and the optimal own fertility differ across types) and $v_j(\phi)$ captures type-specific continuation values. As is standard, (16) is to be interpreted as a log-linear approximation around a balanced-growth reference point; the residual non-log-linear contribution from $-\psi(n)/w$ is absorbed into the type-specific function $v_j(\phi)$.

Under this conjecture, the first-order condition for fertility (recalling that in a continuum of households the individual choice does not affect aggregate N') is:

$$\frac{\psi'(n_j)}{w - \psi(n_j)} = \beta \mathbb{E}_j \left[V_j(A', N', \phi') \right]. \tag{17}$$

With the quadratic cost $\psi(n) = \frac{\gamma}{2} n^2$, this becomes:

$$\frac{\gamma n_j}{w - \frac{\gamma}{2} n_j^2} = \beta \mathbb{E}_j \left[V_j(A', N', \phi') \right]. \tag{18}$$

The following lemma characterizes equilibrium fertility.

Lemma 1 (Equilibrium Fertility)**.** *Under Assumptions 1–3 and the value function conjecture*

(16), *equilibrium fertility for type* $j \in \{R, M\}$ *satisfies:*

$$n_j^* = n^* \cdot h(\theta_j; \bar{n}), \tag{19}$$

where n^* *is a baseline fertility level, and* $h(\theta; \bar{n})$ *is increasing in* θ *with* $h(\bar{\theta}; \bar{n}) = 1$ *for some reference belief* $\bar{\theta}$*. Moreover,* $n_R^* > n_M^*$ *for all equilibrium values of* $\bar{n}$*.*

The proof is in Appendix A. The key economic insight is that households who believe population enhances productivity (Romerians) perceive a higher return to children and hence choose higher fertility. This prediction—that Malthusian beliefs lead to lower fertility—is testable. I examine empirical support for this prediction in the next section.

3 Empirical Evidence: Beliefs and Fertility

A central prediction of the model is that Malthusian beliefs reduce fertility. Lemma 1 establishes that households believing population diminishes productivity ($\theta_M < 0$) choose lower fertility than households believing population enhances productivity ($\theta_R > 0$). In this section, I examine empirical support for this prediction using survey data on beliefs and fertility. Validating this core mechanism provides a foundation for the strategic interaction and dynamic analyses that follow.

3.1 Data

I use data from the General Social Survey (GSS), a nationally representative survey of U.S. adults conducted by NORC at the University of Chicago. The GSS has been administered since 1972 and includes questions on a wide range of social attitudes and demographic characteristics.

The key belief variable is drawn from the question: "The earth cannot continue to support population growth at its present rate." Respondents indicate their agreement on a five-point scale ranging from "strongly agree" to "strongly disagree." I classify respondents who agree or strongly agree as holding *Malthusian beliefs*, corresponding to the view that population growth is unsustainable. This question was asked in the 2000 and 2010 survey waves.

I examine two fertility measures. The first is stated preferences: respondents' answer to "What do you think is the ideal number of children for a family to have?" Responses range from zero to seven or more. The second is realized fertility: the actual number of children the respondent has had. Examining both measures allows me to test whether beliefs shape not only fertility intentions but also fertility outcomes.

I restrict the sample to respondents with valid responses to both the belief and fertility questions, as well as to demographic controls including age, education, and sex. The resulting sample contains 1,822 observations.

3.2 Descriptive Patterns

Table 1 presents summary statistics. The mean ideal number of children is 2.50, with substantial variation (standard deviation 0.87). The mean actual number of children is 1.81 (standard deviation 1.64), reflecting that many respondents have not yet completed their fertility. Approximately 55 percent of respondents hold Malthusian beliefs (agree or strongly agree that Earth cannot support continued population growth). The sample is 56 percent female, with mean age of 47 and mean education of 13.3 years.

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from General Social Survey, 2000 and 2010 waves. Malthusian belief (binary) equals one if respondent agrees or strongly agrees that Earth cannot continue to support population growth. Malthusian belief (1–5 scale) is reverse-coded so that higher values indicate stronger Malthusian beliefs.

Figure 1 displays mean ideal and actual fertility by the intensity of Malthusian beliefs. The pattern is striking: there is a monotonic negative relationship between Malthusian belief strength and both fertility measures. Respondents who strongly disagree that Earth cannot support population growth (the least Malthusian) report an average ideal of 2.98 children and have 1.76 children, while those who strongly agree (the most Malthusian) report an average ideal of 2.24 children and have 1.52 children. The gradient is present for both stated preferences and realized outcomes, though the relationship is stronger for ideal fertility.

A potential concern is that this relationship is driven by education: more educated individuals may both hold more Malthusian beliefs (through exposure to environmental discourse) and desire fewer children (through higher opportunity costs). Figure 2 addresses this concern by displaying mean ideal fertility separately by education group and belief type.

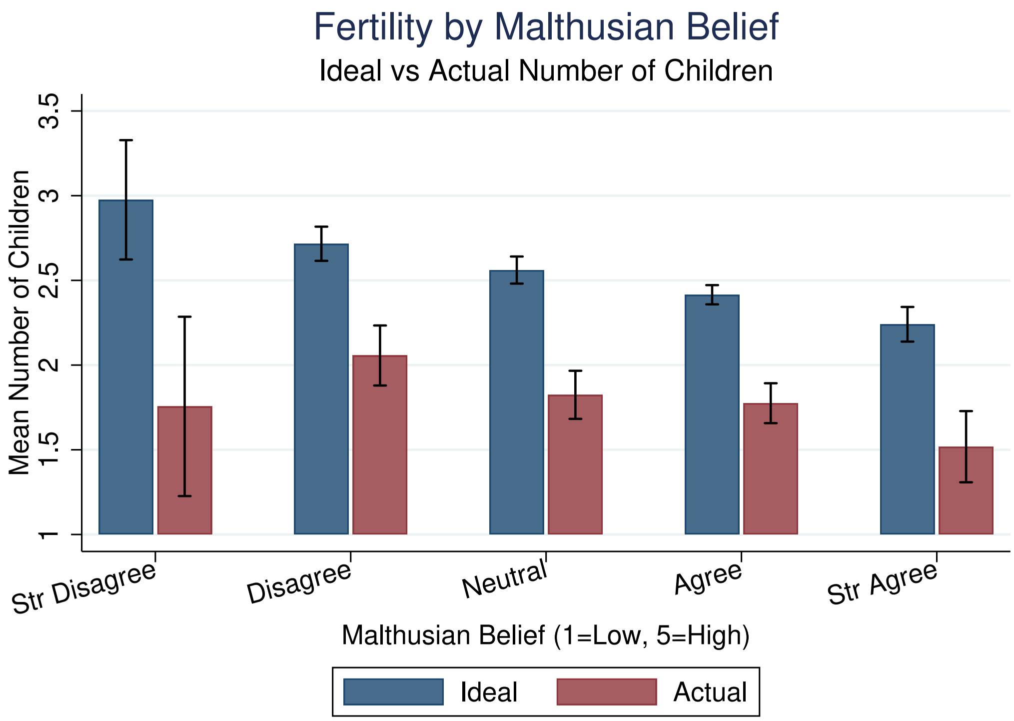

Figure 1. Fertility by Malthusian Belief Intensity

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal (navy) and actual (maroon) number of children for each level of the Malthusian belief scale, where 1 = "strongly disagree" and 5 = "strongly agree" that Earth cannot continue to support population growth. Error bars indicate 95% confidence intervals.

Within every education category—less than high school, high school graduate, some college, and college or more—Malthusians report lower ideal fertility than non-Malthusians. The gap ranges from 0.20 children among college graduates to 0.45 children among those with less than high school education. This pattern suggests that the belief-fertility relationship is not simply a proxy for education.

3.3 Regression Analysis

To assess the robustness of these patterns, I estimate linear regressions of both ideal and actual fertility on Malthusian beliefs with demographic controls. Table 2 presents the results.

Columns (1) and (2) report results for ideal fertility. Malthusians report ideal fertility 0.27 children lower than non-Malthusians (column 1), statistically significant at the 1 percent level. This represents approximately 11 percent of the sample mean. Using the continuous belief scale (column 2), each one-unit increase in Malthusian belief intensity is associated with 0.15 fewer ideal children.

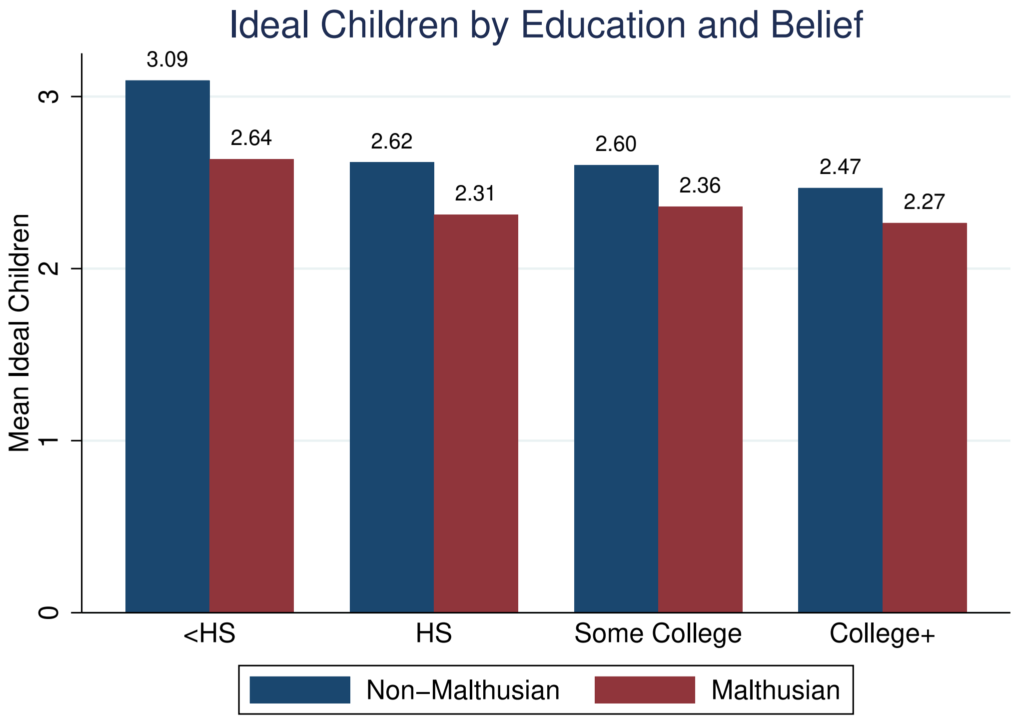

Figure 2. Ideal Number of Children by Education and Malthusian Belief

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal number of children for non-Malthusians (navy) and Malthusians (maroon) within each education category. Malthusian is defined as agreeing or strongly agreeing that Earth cannot continue to support population growth. Numbers above bars indicate group means.

Columns (3) and (4) report results for actual fertility. The coefficient on Malthusian beliefs is -0.29 (column 3), implying that Malthusians have approximately 0.29 fewer children than non-Malthusians—a difference of 16 percent relative to the sample mean of 1.81. The effect using the continuous scale is -0.15 per unit (column 4). Notably, the coefficient on actual fertility is *larger* in absolute value than the coefficient on ideal fertility, suggesting that beliefs translate into behavior with at least as much force as they affect stated preferences.

The higher R^2 for actual fertility (0.26 versus 0.06) reflects the strong relationship between age and realized fertility: older respondents have had more time to accumulate children. The coefficient on age for actual fertility is positive (0.11 per year), while for ideal fertility it is slightly negative (-0.01), consistent with younger cohorts expressing somewhat higher fertility ideals.

Table 2. Effect of Malthusian Beliefs on Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	-0.268^{***}		-0.292^{***}	
	(0.041)		(0.066)	
Malthusian (continuous)		-0.153^{***}		-0.146^{***}
		(0.021)		(0.033)
Age	-0.014^{**}	-0.015^{**}	0.112^{***}	0.111^{***}
	(0.006)	(0.006)	(0.010)	(0.010)
Female	0.015	0.026	0.091	0.104
	(0.040)	(0.040)	(0.068)	(0.068)
Education	-0.047^{***}	-0.046^{***}	-0.144^{***}	-0.143^{***}
	(0.007)	(0.007)	(0.013)	(0.013)
Year FE	Yes	Yes	Yes	Yes
Observations	1,822	1,822	1,822	1,822
R^2	0.060	0.069	0.262	0.262

Notes: All models include age squared and year fixed effects. Malthusian is a binary indicator for agreeing or strongly agreeing that Earth cannot continue to support population growth. Malthusian (continuous) is a 1–5 scale where higher values indicate stronger Malthusian beliefs. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.4 Completed Fertility

A concern with analyzing actual fertility is that younger respondents may not have completed their childbearing. I address this by restricting the sample to respondents with likely completed fertility: women aged 45 and older, and men aged 50 and older. Table 3 presents results for this subsample.

Among those with completed fertility (column 2), Malthusians have 0.26 fewer children than non-Malthusians, significant at the 5 percent level. The coefficient is similar in magnitude to the full-sample estimate, confirming that the relationship holds for realized lifetime fertility, not just current fertility among a mix of ages. The relationship is also present among those who have not yet completed fertility (column 3), with a coefficient of -0.29, suggesting that belief-driven fertility differentials emerge early in the reproductive life course.

3.5 Heterogeneity

Table 4 examines whether the belief-fertility relationship varies across demographic groups. I estimate the full specification separately by education level and sex for both fertility mea-

Table 3. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	-0.292^{***}	-0.257^{**}	-0.288^{***}
	(0.066)	(0.109)	(0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

Notes: Dependent variable is actual number of children. All models control for age, age squared, sex, education, and year. Completed fertility defined as women age 45+ or men age 50+. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

sures.

Table 4. Heterogeneity in the Belief-Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	-0.318^{***}	-0.235^{***}	-0.209^{*}	-0.392^{***}
	(0.069)	(0.049)	(0.115)	(0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: Dependent variable is ideal (columns 1–2) or actual (columns 3–4) number of children. All models control for age, age squared, sex, and year. Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The belief-fertility relationship is present across education groups for both outcome measures. For ideal fertility, the coefficient is somewhat larger for less educated respondents (-0.32) than for more educated respondents (-0.24). For actual fertility, the pattern reverses: the coefficient is larger for more educated respondents (-0.39) than for less educated respondents (-0.21). This suggests that while Malthusian beliefs affect fertility preferences across the education distribution, the translation of beliefs into behavior may be stronger among the more educated, perhaps because they face fewer constraints on achieving their fertility targets.

3.6 Interpretation and Limitations

The empirical findings are consistent with the model's central prediction: individuals who believe population growth is unsustainable (Malthusians) desire and have fewer children than those who do not hold such beliefs. The relationships are economically meaningful—gaps of 0.27 ideal children (11 percent) and 0.29 actual children (16 percent)—and statistically robust across specifications.

Two features of the results merit emphasis. First, the relationship holds for *actual* fertility, not just stated preferences. This addresses a common concern that surveys capture cheap talk rather than consequential behavior. The fact that Malthusians have fewer children, even among those with completed fertility, suggests that beliefs do translate into fertility outcomes.

Second, the coefficient on actual fertility is at least as large as the coefficient on ideal fertility. This is notable because one might expect beliefs to affect intentions more strongly than behavior, given the many constraints (partner preferences, income, fecundity) that intervene between preferences and outcomes. The finding that the behavioral effect is similar or larger suggests either that beliefs operate partly through channels beyond conscious intention, or that individuals with stronger beliefs are more committed to acting on them.

Several limitations warrant emphasis. The evidence remains correlational. The documented association could reflect reverse causality (individuals who have or want fewer children rationalize this by adopting Malthusian beliefs) or omitted variable bias (unobserved characteristics affecting both beliefs and fertility). The cross-sectional nature of the data precludes tracking how belief changes affect subsequent fertility decisions.

Additionally, the belief measure captures concern about global population sustainability, not specifically the scale-effect parameter θ in the model. Respondents may agree that Earth cannot support population growth for reasons unrelated to productivity—for instance, concerns about biodiversity, quality of life, or resource exhaustion—that do not map directly onto the Romerian-Malthusian dichotomy.

Despite these limitations, the evidence provides suggestive support for the belief channel emphasized in the model. The robust correlations between Malthusian beliefs and both ideal and actual fertility, documented across demographic groups and specifications, are what the model predicts. More definitive causal identification remains an important direction for future research. With this empirical foundation established, I now turn to the strategic implications of belief heterogeneity.

4 Strategic Complementarity and Substitutability

I now analyze strategic interaction in fertility choices. The key question is: how does a household's optimal fertility respond to expectations about aggregate fertility? I show that, under empirically plausible parameter conditions, the answer differs by type: Romerians exhibit strategic complementarity while Malthusians exhibit strategic substitutability. This asymmetry has important implications for equilibrium properties.

4.1 The Fertility Game

Consider the fertility choice as a game among households. Each household i chooses n_i to maximize expected utility. Aggregate fertility is $\bar{n} = \int n_i \, di$. The household's payoff depends on $(n_i, \bar{n})$ through the following channel: higher $\bar{n}$ raises $N' = \bar{n} \cdot N$, which affects A'' through (2) (and hence subsequent productivity) and dilutes future per capita consumption.

To formalize, let $U_j(n_j, \bar{n})$ denote the expected utility of a type-j household choosing fertility n_j when aggregate fertility is $\bar{n}$. This expected utility accounts for future consumption, which depends on future TFP and population under the household's subjective belief θ_j.

Definition 2 (Strategic Complementarity/Substitutability)**.** *Household type j exhibits* strategic complementarity *if* $\frac{\partial^2 U_j}{\partial n_j \partial \bar{n}} > 0$ *and* strategic substitutability *if* $\frac{\partial^2 U_j}{\partial n_j \partial \bar{n}} < 0$.

Strategic complementarity means that the marginal return to fertility is increasing in aggregate fertility. Strategic substitutability means the opposite.

4.2 Analysis Using Global Games Techniques

To analyze strategic interaction rigorously, I adapt the global games framework of Carlsson and Van Damme (1993) and Morris and Shin (1998). The standard global games setup features a continuum of players with private signals about a fundamental, facing a binary action choice in a game with strategic complementarities. I extend this to allow for continuous actions (fertility) and for heterogeneous strategic interaction (complementarity for Romerians, substitutability for Malthusians).

Setup. Suppose the economy is in a state where current TFP A and population N are known, and the Romerian share is ϕ. Each household knows its own type (and hence belief θ_j) but faces uncertainty about the types of others and hence about aggregate fertility $\bar{n}$.

Let $n_j(\bar{n}^e)$ denote the fertility choice of type j when the expected aggregate fertility is $\bar{n}^e$. The actual aggregate fertility is then:

$$\bar{n} = \phi \cdot n_R(\bar{n}^e) + (1 - \phi) \cdot n_M(\bar{n}^e). \tag{20}$$

A rational expectations equilibrium requires $\bar{n}^e = \bar{n}$: expected aggregate fertility equals actual aggregate fertility.

Best Response Functions. To derive best response functions, I compute how each type's optimal fertility depends on expected aggregate fertility. Recall that current-period TFP growth depends on *current* population N_t, not on current aggregate fertility $\bar{n}_t$. Aggregate fertility today, $\bar{n}_t$, affects future variables only through $N_{t+1} = \bar{n}_t N_t$, which then influences A_{t+1} (under non-zero scale effects) and per capita consumption.

Using the value function conjecture, the continuation value summarizes both effects. Under belief θ_j, the expected continuation value $\mathbb{E}_j[V_j(A', N', \phi')]$ is increasing in $\bar{n}^e$ precisely when the discounted scale-effect benefit outweighs the per capita dilution penalty. Standard recursive calculation (carried out in the Appendix) shows that the marginal effect has the same sign as

$$(\alpha - 1)(1 - \beta n_j) + \beta n_j \theta_j. \tag{21}$$

This sign is unambiguously negative for Malthusians (since $\theta_j < 0$ and $\alpha - 1 < 0$, both summands are negative). For Romerians, it is positive when θ_R is sufficiently large, specifically when

$$\theta_R > \frac{(1-\alpha)(1-\beta n_R)}{\beta n_R}. \tag{22}$$

For empirically plausible values—$\alpha \approx 0.7$, $\beta \approx 0.95$, $n_R \approx 1$—the right-hand side is on the order of 0.016, which is comfortably below the calibrated $\theta_R \approx 0.1$ used later. Throughout the rest of the analysis, I maintain the standing assumption that (22) holds.

Proposition 1 (Strategic Interaction)**.** *Consider the fertility game under Assumptions 1–3. Suppose condition (22) holds. For type $j \in \{R, M\}$, let $n_j^*(\bar{n}^e)$ denote the optimal fertility given expected aggregate fertility $\bar{n}^e$. Then:*

(i) Romerians exhibit strategic complementarity: $\frac{dn_R^*}{d\bar{n}^e} > 0$.

(ii) Malthusians exhibit strategic substitutability: $\frac{dn_M^*}{d\bar{n}^e} < 0$.

The proof is in Appendix A. The intuition is as follows. When a Romerian expects higher aggregate fertility, they expect higher future population, which under Romerian beliefs translates to faster productivity growth in subsequent generations. Provided this benefit dominates the dilution of per capita consumption (condition (22)), the continuation value of each child rises, increasing the marginal benefit of fertility. The Romerian responds by having more children. When a Malthusian expects higher aggregate fertility, both forces—slower future productivity growth and dilution of per capita consumption—work in the same direction, lowering the continuation value and reducing fertility.

4.3 Equilibrium Characterization

Proposition 1 implies that the aggregate best response function—the mapping from expected aggregate fertility to actual aggregate fertility—depends on the composition of the population. Defining $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1-\phi)n_M^*(\bar{n}^e)$, the slope satisfies

$$\frac{dBR}{d\bar{n}^e} = \phi \frac{dn_R^*}{d\bar{n}^e} + (1-\phi)\frac{dn_M^*}{d\bar{n}^e}.$$

This slope is positive when ϕ is sufficiently large (Romerian dominance), negative when ϕ is sufficiently small (Malthusian dominance), and zero at a critical ϕ^* where the two effects exactly offset (a formal statement and proof are deferred to Corollary 1 in Appendix A). The critical share ϕ^* satisfies:

$$\phi^* = \frac{|dn_M^*/d\bar{n}^e|}{|dn_R^*/d\bar{n}^e| + |dn_M^*/d\bar{n}^e|}. \tag{23}$$

When $\phi > \phi^*$, the aggregate best response function slopes upward, creating the potential for multiple equilibria. A slight increase in expected fertility raises actual fertility, which can be self-fulfilling. When $\phi < \phi^*$, the aggregate best response slopes downward, ensuring a unique equilibrium: any deviation from equilibrium is self-correcting.

Proposition 2 (Equilibrium Multiplicity)**.** *Under the conditions of Proposition 1:*

(i) If $\phi < \phi^*$, there exists a unique equilibrium.

(ii) If $\phi > \phi^*$ and the slope of $BR(\bar{n}^e)$ exceeds 1 at some point, there exist multiple equilibria.

In the multiple-equilibria region, equilibria can be ranked by aggregate fertility: there exists a low-fertility equilibrium, a high-fertility equilibrium, and potentially unstable intermediate equilibria.

The proof follows standard arguments from the global games literature and is provided in Appendix A. The economic implication is that societies dominated by Romerians may be subject to self-fulfilling fluctuations in fertility, whereas societies dominated by Malthusians have more stable fertility dynamics.

4.4 Uniqueness under Incomplete Information

A key insight from the global games literature is that introducing small amounts of private information can select a unique equilibrium even when multiple equilibria exist under common knowledge. Suppose that each household observes a private signal about the Romerian share, $\phi_i = \phi + \eta_i$, with $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$ i.i.d. across households. Households must form expectations about ϕ (and hence about aggregate fertility) based on their private signal and prior beliefs.

In Appendix A (Proposition 6), I adapt the standard global-games selection argument of Morris and Shin (1998) to show that, in the multiple-equilibria region of Proposition 2, as signal noise $\sigma_\eta \to 0$ there exists a unique equilibrium of the incomplete-information game in which each type uses a threshold strategy on its private signal. Romerians coordinate on the high-fertility equilibrium when their signal exceeds a common threshold ϕ^{**} and on the low-fertility equilibrium otherwise; Malthusians, because of strategic substitutability, choose their low-fertility response when the signal exceeds ϕ^{**} and their high-fertility response otherwise. The threshold ϕ^{**} depends on the relative strength of strategic complementarity among Romerians and strategic substitutability among Malthusians: when Romerian strategic complementarity is strong, the threshold is lower, meaning that Romerians coordinate on high fertility even when the Romerian share is moderate.

The key economic insight is that private information breaks the common-knowledge structure that supports multiple equilibria. Each household is uncertain about whether other households are above or below the threshold, and this uncertainty pins down a unique equilibrium.

5 Long-Run Dynamics and the Role of Wrong Beliefs

I now analyze the long-run dynamics of beliefs and population. The central insight is that wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. In Romerian worlds, Malthusian minorities cause prolonged stagnation through an informational poverty trap. In Malthusian worlds, Romerian minorities cause boom-bust cycles. This asymmetry arises from the interaction between demographic selection (which always favors optimists) and Bayesian learning (which favors correct beliefs, but only when signals are informative).

5.1 Two Forces: Demographic Selection and Learning

The Romerian share ϕ_t evolves according to (14):

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda)\pi_{t+1}. \tag{24}$$

Two forces drive this evolution.

Demographic selection. The term $\phi_t^{birth} = \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t) n_M}$ captures the effect of differential fertility. Since $n_R > n_M$ (Romerians have more children), we have $\phi_t^{birth} > \phi_t$ for any $\phi_t \in (0, 1)$. Demographic selection pushes ϕ_{t+1} upward regardless of which belief is correct.

Bayesian learning. The term π_{t+1} captures the effect of evidence. The direction depends on the true θ^*: if $\theta^* > 0$ (Romerian world), high population tends to generate high productivity growth, which is evidence for the Romerian model, so π_{t+1} tends to exceed π_t in expectation; if $\theta^* < 0$ (Malthusian world), high population tends to generate low productivity growth, which is evidence for the Malthusian model, so π_{t+1} tends to fall below π_t in expectation.

Crucially, the *strength* of learning depends on population through the signal informativeness $\mathcal{I}_t = (\theta_R - \theta_M)^2 (\ln N_t)^2 / (2\sigma_\varepsilon^2)$. When $N_t \approx 1$, learning is weak regardless of the true θ^*.

5.2 Dynamics in a Romerian World: Convergence and the Informational Poverty Trap

I first establish that, in a Romerian world, beliefs eventually converge to the truth.

Theorem 1 (Convergence in Romerian Worlds)**.** *Suppose $\theta^* > 0$. Then, starting from any initial condition (ϕ_0, N_0, A_0) with $\phi_0 \in (0, 1)$, the economy converges almost surely to a steady state with:*

(i) $\phi_t \to 1$: Romerians dominate the population.

(ii) $\pi_t \to 1$: posterior probability of Romerian world converges to certainty.

(iii) Aggregate fertility converges to n_R^*.

(iv) Population grows at rate $n_R^* - 1$ in the long run.

The proof is in Appendix A. The intuition is that both demographic selection and learning favor Romerians when the world is truly Romerian. Demographic selection favors them because they have more children. Learning favors them because observed outcomes confirm their beliefs.

However, Theorem 1 is a qualitative result about almost-sure convergence. It is silent on the *rate* of convergence. The following analysis shows that convergence can be arbitrarily slow when the economy starts in the "informational poverty trap" region.

The Uninformative Region.

Definition 3 (Uninformative Region)**.** *For $\delta > 0$, define the* δ-uninformative region *as the set of population levels satisfying $|\ln N_t| < \delta$. In this region, the per-period expected information gain satisfies:*

$$\mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 (\ln N_t)^2}{2\sigma_\varepsilon^2} < \frac{(\theta_R - \theta_M)^2 \delta^2}{2\sigma_\varepsilon^2} \equiv \bar{\mathcal{I}}(\delta). \tag{25}$$

When δ is small, $\bar{\mathcal{I}}(\delta)$ is small, and signals reveal almost nothing about the true θ^*.

The Informational Poverty Trap. The informational poverty trap arises when pessimistic beliefs keep the economy in the uninformative region, and the uninformative region prevents beliefs from being corrected.

Definition 4 (Trap Region)**.** *The* informational poverty trap region $\mathcal{T}(\delta, \eta)$ *is the set of states (ϕ, N) such that: (i) population is in the δ-uninformative region, $|\ln N| < \delta$; and (ii) aggregate fertility is near replacement, $\bar{n}(\phi) \equiv \phi n_R + (1-\phi) n_M \in [1, 1+\eta]$ for small $\eta > 0$.*

In the trap region, population grows slowly (if at all), so the economy remains in the uninformative region. Learning is weak, so beliefs evolve slowly. The trap is self-reinforcing.

The following assumption ensures the trap region is non-empty.

Assumption 4 (Fertility Straddles Replacement)**.** *Malthusian fertility is below replacement and Romerian fertility is above replacement: $n_M < 1 < n_R$.*

Under Assumption 4, there exists a critical Romerian share ϕ^{trap} such that $\bar{n}(\phi^{trap}) = 1$:

$$\phi^{trap} = \frac{1 - n_M}{n_R - n_M}. \tag{26}$$

For ϕ slightly above ϕ^{trap}, aggregate fertility slightly exceeds 1, and population grows slowly.

Slow Escape from the Trap. The following result characterizes escape time from the trap region.

Proposition 3 (Slow Learning in the Trap)**.** *Consider a Romerian world ($\theta^* = \theta_R > 0$) under Assumption 4. Suppose the initial state satisfies $(\phi_0, N_0) \in \mathcal{T}(\delta, \eta)$ for small $\delta, \eta > 0$. Define the escape time as:*

$$T^{esc} = \inf\{t : \phi_t > \bar{\phi}\}, \tag{27}$$

for some target $\bar{\phi} > \phi_0$. Then:

(i) The per-period expected change in the Romerian share from learning is bounded by:

$$\mathbb{E}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t, \textit{learning}] \leq C_1 \cdot \bar{\mathcal{I}}(\delta), \tag{28}$$

for a constant $C_1 > 0$ depending on model parameters.

(ii) The per-period expected change from demographic selection is bounded by:

$$\phi_{t+1}^{birth} - \phi_t \leq C_2 \cdot (n_R - n_M) \cdot (1 - \phi_t), \tag{29}$$

for a constant $C_2 > 0$.

(iii) The expected escape time satisfies:

$$\mathbb{E}[T^{esc}] \geq \frac{\bar{\phi} - \phi_0}{\lambda C_2(n_R - n_M) + (1-\lambda) C_1 \bar{\mathcal{I}}(\delta)}. \tag{30}$$

The proof is in Appendix A. The key implication is that as $\delta \to 0$ (population closer to 1), the learning contribution $(1-\lambda)C_1\bar{\mathcal{I}}(\delta) \to 0$, and escape relies entirely on demographic selection. But demographic selection is slow when the fertility gap $n_R - n_M$ is small or when cultural transmission λ is weak. As an immediate consequence (Corollary 2 in Appendix A), the expected escape time diverges as $\Omega(1/\delta^2)$ and can exceed any finite horizon for sufficiently small δ. Thus, although Theorem 1 guarantees eventual escape, escape can take arbitrarily long when the economy starts with population near 1 and pessimistic beliefs.

To understand when the economy is trapped versus when it escapes quickly, Proposition 7 in Appendix A characterizes the boundary of the trap region in terms of primitives. The critical population beyond which learning dominates demographic selection is

$$N^{crit}(\phi) = \exp\left(\sqrt{\frac{2\sigma_\varepsilon^2 \cdot \lambda C_2(n_R - n_M)(1-\phi)}{(1-\lambda)C_1(\theta_R - \theta_M)^2}}\right).$$

The trap boundary depends on the ratio of demographic selection strength (numerator) to learning informativeness (denominator). High noise σ_ε^2 raises N^{crit}, making the trap larger. Strong cultural transmission (λ high) also raises N^{crit}, because it reduces reliance on learning. Large belief dispersion $(\theta_R - \theta_M)^2$ lowers N^{crit}, because it makes signals more diagnostic when population does deviate from 1.

Summary: Wrong Beliefs in Romerian Worlds. In a Romerian world, Malthusian minorities cause prolonged stagnation through the informational poverty trap. The trap is self-reinforcing: pessimism suppresses fertility, low fertility keeps population in the uninformative region, uninformative signals fail to correct pessimism. Escape eventually occurs, but can take arbitrarily long. The welfare cost of Malthusian beliefs in a Romerian world is not merely slower convergence—it is the possibility of missing opportunities for growth for extended periods, potentially exceeding historical timescales.

5.3 Dynamics in a Malthusian World: Cycles

The dynamics are starkly different when $\theta^* < 0$.

Theorem 2 (Cycles in Malthusian Worlds)**.** *Suppose $\theta^* < 0$. Then, for intermediate values of the learning responsiveness (parameterized by σ_ε^{-2}), the linearized dynamics around an interior steady state admit complex eigenvalues with modulus exceeding one, generating local oscillations. In the nonlinear system, these oscillations are bounded and produce endogenous cycles in which:*

(i) The Romerian share ϕ_t oscillates, never converging to 0 or 1.

(ii) Population N_t oscillates, with booms and busts.

(iii) Per capita consumption c_t oscillates countercyclically with ϕ_t.

The cycle period depends on the transmission parameter λ and the learning responsiveness.

The proof is in Appendix A. The key mechanism is the conflict between demographic selection and learning.

Cycle dynamics. Starting from an interior ϕ_0:

Expansion phase: Suppose population is moderate and welfare is acceptable. Both models fit the data reasonably well, so learning is weak. Demographic selection dominates: $\phi_{t+1} > \phi_t$ because Romerians have more children. Rising ϕ raises aggregate fertility, causing population to grow.

Crisis phase: As population grows, productivity growth slows (since $\theta^* < 0$). Per capita consumption falls. The Malthusian model fits the data much better than the Romerian model: high population coincided with poor outcomes. Young generations update sharply toward Malthusian beliefs: π_{t+1} falls. If learning is sufficiently responsive, this overwhelms demographic selection, and $\phi_{t+1} < \phi_t$.

Recovery phase: Falling ϕ reduces aggregate fertility. Population growth slows or reverses. Per capita consumption stabilizes and begins recovering. With moderate population, both models again fit the data similarly, weakening learning. Demographic selection again dominates, and ϕ begins rising, starting the next cycle.

Conditions for cycles. Not all Malthusian worlds exhibit cycles. If learning is very weak (high σ_ε), demographic selection always dominates, and ϕ increases monotonically toward 1. The economy converges to a steady state with Romerian beliefs, even though these beliefs are incorrect. Population is inefficiently high, and welfare is inefficiently low, but there are no cycles.

If learning is very strong (low σ_ε), young generations quickly identify the true model, and Malthusian beliefs spread rapidly. The interior steady state is destabilized in favor of a corner steady state with Malthusian dominance, low fertility, and stable population.

Cycles occur for intermediate learning responsiveness, where demographic selection and learning are both operative but neither dominates and the interior steady state is locally an unstable focus. A formal characterization of the cycle existence region in terms of an index of learning responsiveness and an index of demographic selection strength is provided by Proposition 8 in Appendix A.

5.4 The Asymmetry: Why Wrong Beliefs Generate Different Pathologies

The analysis reveals a fundamental asymmetry between the two scenarios, summarized in Table 5.

Table 5. Pathologies from Wrong Beliefs

	Romerian World ($\theta^* > 0$)	Malthusian World ($\theta^* < 0$)
Wrong belief	Malthusian ($\theta_M < 0$)	Romerian ($\theta_R > 0$)
Demographic selection	Favors Romerians (correct)	Favors Romerians (incorrect)
Learning	Favors Romerians (correct)	Favors Malthusians (correct)
Forces align?	Yes	No
Pathology	Informational poverty trap	Boom-bust cycles
Long-run outcome	Convergence to truth	Cycles or incorrect steady state

In Romerian worlds, both forces align toward correct beliefs. Demographic selection favors optimists, and optimists happen to be right. Learning favors the correct model, which is optimistic. Incorrect (Malthusian) beliefs are self-correcting, but the correction can be very slow if the economy starts in the trap.

In Malthusian worlds, the forces conflict. Demographic selection favors optimists, but optimists are wrong. Learning favors the correct model, which is pessimistic. The result is either persistent incorrect beliefs (if demographic selection dominates) or cycles (if neither force dominates).

This asymmetry has a simple but profound implication: *human societies are more vulnerable to excessive optimism than excessive pessimism in terms of stability, but more vul-*

nerable to excessive pessimism in terms of stagnation. Optimists reproduce more, regardless of whether their optimism is warranted, which destabilizes Malthusian worlds. Pessimism suppresses population and thereby suppresses learning, which traps Romerian worlds.

Remark 1 (The Core Insight)**.** *The central theoretical contribution can be stated sharply:* in a Romerian world, Malthusians cause stagnation; in a Malthusian world, Romerians cause instability*. Wrong beliefs are always costly, but the* nature *of the cost depends on the direction of the error. This asymmetry arises because demographic selection always favors optimists, while learning favors truth—and these forces align or conflict depending on what the truth is.*

6 Demographic History: Two Interpretations

The model provides a unified framework for interpreting broad patterns in demographic history. Crucially, it admits two fundamentally different interpretations of the pre-industrial era, with distinct implications for understanding the modern world.

6.1 Interpretation 1: A Genuinely Malthusian Past

The conventional interpretation, consistent with economic historians from Malthus himself to Clark (2007), is that the pre-industrial world was genuinely Malthusian: scale effects were negative, and larger populations did reduce living standards.

Under this interpretation, the cycles documented by Voigtländer and Voth (2013) and others—periods of demographic expansion followed by crises that reduced population—reflect the dynamics of Theorem 2. During periods of relative prosperity, Romerian-like optimism spread through demographic selection. Population grew. Eventually, population exceeded carrying capacity, living standards fell, and Malthusian beliefs were reinforced by crisis. Population stabilized or declined, recovery began, and the cycle restarted.

The Black Death of 1347–1351 provides a particularly clear example. Europe's population had grown substantially in the preceding centuries, straining resources. The plague killed roughly one-third of the population. Real wages rose sharply in the aftermath and remained elevated for over a century. The model interprets this as a crisis phase that both directly reduced population and reinforced Malthusian beliefs, leading to a sustained period of lower fertility.

Under this interpretation, the Industrial Revolution marked a genuine regime change from $\theta^* < 0$ to $\theta^* > 0$. Sustained technological progress, enabled by the accumulation of scientific knowledge and the development of institutions supporting innovation, broke the

Malthusian link between population and living standards. The world shifted from being truly Malthusian to truly Romerian.

6.2 Interpretation 2: An Always-Romerian World Trapped by Information Poverty

The informational poverty trap opens a radically different interpretation: the world may have *always* been Romerian, but humanity was trapped for millennia by the self-reinforcing dynamics of pessimism and low population.

Under this interpretation, positive scale effects existed throughout history—more people always meant more ideas, more specialization, more innovation. But populations were too small, and productivity signals too noisy, for anyone to reliably learn this fact. Malthusian beliefs, once established, suppressed fertility, keeping population low. Low population generated uninformative signals. Uninformative signals failed to correct Malthusian beliefs. The trap persisted not because Malthusianism was correct, but because the evidence needed to refute it could only be generated by escaping it.

The observed pre-industrial cycles, under this interpretation, were not the inevitable boom-bust dynamics of a Malthusian world but rather failed escape attempts from the trap. Occasionally, some region or era experienced conditions that pushed population higher—better weather, new crops, reduced warfare. Beliefs began to update toward optimism. But the gains were fragile: any negative shock (plague, famine, invasion) was interpreted as confirmation of Malthusian fears, beliefs snapped back, fertility fell, and the economy returned to the trap.

The Industrial Revolution, under this interpretation, was not a technological regime change but an *informational escape*. Some combination of factors—the Enlightenment, improving institutions, a sequence of positive productivity shocks—pushed European populations above the critical threshold N^{crit}. Once in the informative region, learning accelerated. Evidence of positive scale effects accumulated. Beliefs shifted decisively toward Romerianism. Fertility rose further. Population exploded. The trap dissolved, not because the world changed, but because humanity finally accumulated enough information to see what had always been true.

6.3 Distinguishing the Interpretations

Both interpretations are consistent with the qualitative features of demographic history: long periods of stagnation, occasional expansions, crises, and eventual escape to modern growth. Distinguishing them requires careful attention to details of the historical record and to quantitative magnitudes.

Several features favor the genuine-regime-change interpretation. First, the timing of the Industrial Revolution coincides with specific technological innovations (steam power, mechanization) that plausibly altered the relationship between population and productivity. Second, the scale of technological progress after 1800 suggests a qualitative break rather than just informational updating. Third, environmental constraints (land, resources) appear to have genuinely bound pre-industrial economies in ways that seem inconsistent with large positive scale effects.

Several features favor the informational-escape interpretation. First, there is evidence of positive scale effects even in pre-industrial settings: larger cities were more innovative, more connected regions were more prosperous. Second, the speed of the transition, once it began, is more consistent with belief updating than with gradual technological change. Third, the modern fertility decline in developed countries—discussed below—is puzzling if the world is truly Romerian but explicable if beliefs can drift back toward Malthusianism even in a Romerian world.

I do not take a definitive stand on which interpretation is correct. The contribution of the model is to show that *both are theoretically coherent* and that they have *different implications for policy*. If the first interpretation is correct, the key question for today is whether the world might shift back to Malthusian due to environmental constraints or automation. If the second interpretation is correct, the key question is how to prevent beliefs from drifting back into a trap even though the underlying fundamentals remain Romerian.

6.4 The Modern Fertility Decline

Fertility rates in developed countries have fallen dramatically since the 1960s, reaching levels well below replacement in most of Europe and East Asia. This decline is puzzling from a simple Romerian perspective: if larger populations are beneficial, why are the most prosperous societies choosing the fewest children?

The model suggests several interpretations.

Revival of Malthusian beliefs. The environmental movement, which emerged prominently in the 1960s and 1970s, can be understood as a revival of Malthusian thinking. Concerns about pollution, resource depletion, climate change, and ecological limits represent beliefs that population growth is harmful. The empirical evidence in Section 3 documents that such beliefs are prevalent—over half of GSS respondents agree that Earth cannot support continued population growth—and are associated with substantially lower fertility. Under the informational-escape interpretation, the modern fertility decline represents a partial return toward the trap: Malthusian beliefs are spreading again, suppressing fertility, which may eventually push the economy back toward the uninformative region.

Actual shift to Malthusian technology. Under the regime-change interpretation, the true θ^* may have recently shifted from positive to negative. Automation and artificial intelligence may have reduced the marginal contribution of human labor to economic output. Environmental constraints, particularly climate change, may have imposed genuine costs on population growth. If the world has become Malthusian again, low fertility may be an appropriate response rather than a mistake.

Quantity-quality substitution. The model abstracts from human capital, but in reality, parents face a quantity-quality tradeoff. Rising returns to education may have induced parents to invest more in each child's human capital at the expense of having more children. This is conceptually distinct from Malthusianism but generates similar fertility outcomes.

Distinguishing among these interpretations is empirically challenging but important for policy. If the first interpretation is correct, fertility may be inefficiently low, and policies that correct misperceptions or subsidize childbearing may be warranted. If the second interpretation is correct, low fertility is efficient. If the third interpretation is correct, the efficiency of low fertility depends on whether private returns to human capital equal social returns.

6.5 Contemporary Divergence and Future Dynamics

Fertility today varies enormously across countries and groups. Total fertility rates range from below 1.0 in some East Asian countries to above 6.0 in parts of Sub-Saharan Africa. Within countries, fertility varies systematically with religion, education, and political orientation.

The model predicts that these differences are partially driven by belief heterogeneity and are self-reinforcing through demographic selection. Groups with Romerian beliefs (or with religious or cultural traditions that encourage high fertility for other reasons) will grow in population share over time. If cultural transmission is strong (λ high), these compositional shifts can substantially affect aggregate beliefs and fertility over generations.

This has implications for long-run population projections. Standard demographic projections typically assume convergence to low fertility as countries develop. The model suggests that convergence may be slower or incomplete if belief heterogeneity persists. Groups with high-fertility beliefs may maintain or increase their population share even as mean beliefs shift.

Whether this is efficient depends on the true value of θ^*. If the world remains Romerian, high-fertility groups are moving the population toward efficiency. If the world has become

Malthusian, high-fertility groups are exacerbating the problem. The key empirical question is which technological regime currently prevails.

7 Policy: Welfare Criteria and the Value of Public Information

I now analyze the welfare effects of public announcements about the scale-effect parameter θ. A distinctive feature of this analysis is the careful treatment of welfare evaluation when population is endogenous. As Golosov et al. (2007) emphasize, standard welfare theorems fail when the set of agents depends on allocations. I begin by discussing alternative welfare criteria, then characterize optimal policy and compare decentralized allocations to the planner's solution under each criterion.

7.1 Welfare Criteria with Endogenous Population

Evaluating welfare when fertility is a choice variable raises fundamental conceptual issues. The set of people who exist depends on policy, so we cannot simply ask whether everyone is better off—the "everyone" changes. Following Golosov et al. (2007) and the population ethics literature surveyed by Blackorby et al. (2005), I consider three welfare criteria.

Dynastic Welfare (Respecting Barro-Becker Preferences). The most natural criterion given the model's structure is to respect households' dynastic preferences as stated. Under this criterion, the planner maximizes the expected utility of the initial generation:

$$W^{dyn} = \phi_0 V_R(A_0, N_0, \phi_0) + (1 - \phi_0) V_M(A_0, N_0, \phi_0), \tag{31}$$

where V_j is the value function under belief θ_j, evaluated using households' subjective expectations. This criterion takes preferences as given and does not second-guess whether beliefs are "correct." It corresponds to the P-efficiency concept of Golosov et al. (2007): an allocation is efficient if there is no alternative that makes all existing people better off while creating no additional people.

Paternalistic Welfare (Correcting for Wrong Beliefs). A paternalistic planner might argue that households with incorrect beliefs are making mistakes, and welfare should be evaluated under the true model rather than subjective beliefs. Under this criterion:

$$W^{pat}(\theta^*) = \phi_0 \tilde{V}_R(A_0, N_0, \phi_0; \theta^*) + (1 - \phi_0) \tilde{V}_M(A_0, N_0, \phi_0; \theta^*), \tag{32}$$

where $\tilde{V}_j(\cdot;\theta^*)$ evaluates the outcome of type-j's fertility choice under the *true* law of motion with parameter θ^*, not the household's subjective belief θ_j. This criterion treats incorrect beliefs as causing welfare losses: a Malthusian in a Romerian world is "wrong" and their low fertility is inefficient.

Utilitarian Welfare (Total or Average). Utilitarian criteria aggregate welfare across all individuals who exist. Total utilitarianism maximizes:

$$W^{tot} = \sum_{t=0}^{\infty} \beta^t N_t \cdot u(c_t), \tag{33}$$

the discounted sum of utilities across all people. This criterion favors large populations when $u(c_t) > 0$. Average utilitarianism maximizes:

$$W^{avg} = \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{34}$$

which is neutral about population size and cares only about per-capita welfare.

As Parfit (1984) famously argued, total utilitarianism leads to the "Repugnant Conclusion": a sufficiently large population with barely positive utility is preferred to a small population with very high utility. Critical-level utilitarianism, which adds welfare only when $u(c_t)$ exceeds a threshold $\bar{u}$, avoids this conclusion but requires specifying the threshold.

Remark 2 (Choice of Welfare Criterion)**.** *The appropriate criterion depends on normative priors about population ethics. In what follows, I primarily use paternalistic welfare, which aligns with the paper's treatment of "wrong beliefs" as causing inefficiency. I note where conclusions differ under alternative criteria. The dynastic criterion is most defensible if one respects revealed preference; the utilitarian criteria are most defensible if one gives independent moral weight to potential future people.*

7.2 The First-Best Allocation

Before analyzing announcement policy, I characterize the first-best allocation. This provides a benchmark against which to measure inefficiency.

A planner who knows θ^* and can directly choose fertility solves:

$$\max_{\{n_t\}} \sum_{t=0}^{\infty} \beta^t N_t \cdot u\left(A_t N_t^{\alpha-1} - \psi(n_t)\right) \tag{35}$$

subject to $N_{t+1} = n_t N_t$ and $A_{t+1} = A_t(1 + \bar{g} + \theta^* \ln N_t)$. The key difference from the decentralized problem is that the planner internalizes the effect of fertility on future TFP through the scale effect. The first-order condition for optimal fertility equates the marginal cost of children to the marginal benefit, including the TFP externality:

$$\psi'(n_t^{FB}) = \beta N_t \cdot \mathbb{E}\left[\frac{\partial V^{FB}}{\partial N_{t+1}}\right], \tag{36}$$

where $\frac{\partial V^{FB}}{\partial N_{t+1}}$ includes both the direct effect on future consumption and the indirect effect through TFP growth.

Proposition 4 (First-Best vs. Decentralized Fertility)**.** *In a Romerian world ($\theta^* > 0$):*

(i) The first-best fertility n^{FB} exceeds the decentralized fertility n^{CB} that would obtain if all households shared the correct belief θ^*: $n^{FB} > n^{CB}$.

(ii) Both n^{FB} and n^{CB} exceed Malthusian fertility: $n^{FB} > n^{CB} > n_M^*$.

(iii) The welfare gap between first-best and decentralized equilibrium has two components: an externality gap (present even with common correct beliefs) and a belief-distortion gap (additional loss from heterogeneous beliefs).

In a Malthusian world ($\theta^* < 0$), the ranking reverses for the externality: $n^{FB} < n^{CB}$, as the planner internalizes the negative scale effect.

The proof is in Appendix B. The key insight is that in decentralized equilibrium, households do not internalize the positive (or negative) externality of population on TFP. In a Romerian world, this leads to under-investment in children even when beliefs are correct. Belief heterogeneity adds a second source of inefficiency: Malthusians further reduce fertility below the already-too-low decentralized level. How the first-best fertility differs across welfare criteria—in particular, that under total utilitarianism the planner additionally values having more people, leading to even higher optimal fertility, while under average utilitarianism the optimum is lower—is established formally in Proposition 9 in Appendix B.

7.3 Welfare Decomposition

To understand the sources of inefficiency, I decompose the welfare gap between first-best and heterogeneous-belief equilibrium.

Proposition 5 (Welfare Decomposition)**.** *Under paternalistic welfare, the gap between first-best and heterogeneous-belief equilibrium can be written as:*

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{\Delta_1(\theta^*)}_{\text{externality gap}} + \underbrace{\Delta_2(\theta^*, \bar{\theta})}_{\text{mean belief bias}} + \underbrace{\Delta_3(Var(\theta_i))}_{\text{belief dispersion}}, \tag{37}$$

where:

(i) $\Delta_1(\theta^*) = W^{FB}(\theta^*) - W^{CB}(\theta^*)$ *is the gap between first-best and common-belief decentralized equilibrium—the pure externality loss.*

(ii) $\Delta_2(\theta^*, \bar{\theta}) = W^{CB}(\theta^*) - W^{CB}(\bar{\theta})$ *is the additional loss when mean beliefs* $\bar{\theta} = \phi\theta_R + (1-\phi)\theta_M$ *differ from truth.*

(iii) $\Delta_3 = W^{CB}(\bar{\theta}) - W^{HB} \approx \Omega \cdot Var(\theta_i)$ *is the loss from belief dispersion around the mean, where* $\Omega > 0$ *depends on the curvature of the value function.*

The proof is in Appendix B. This decomposition clarifies that public announcements can potentially reduce Δ_2 (by shifting mean beliefs toward truth) and Δ_3 (by reducing dispersion), but cannot directly address Δ_1 without additional policy instruments such as fertility subsidies.

7.4 The Planner's Announcement Problem

Consider a planner who observes a signal about the true parameter:

$$\hat{\theta} = \theta^* + \nu, \tag{38}$$

where $\nu \sim \mathcal{N}(0, \sigma_\nu^2)$ represents estimation error. The planner can publicly announce a value θ^A. Upon hearing the announcement, households update beliefs according to:

$$\theta_j^{post} = \frac{\tau_j\theta_j + \tau_A\theta^A}{\tau_j + \tau_A}, \tag{39}$$

where τ_j is the precision of household j's prior and τ_A is the credibility of the announcement.

Assumption 5 (Announcement Credibility)**.** *Households treat the public announcement as a signal with precision* $\tau_A > 0$*. When* $\tau_A \to \infty$*, households fully adopt the announced value; when* $\tau_A \to 0$*, households ignore the announcement.*

7.5 Optimal Announcement under Different Criteria

The optimal announcement policy differs substantially across welfare criteria.

Theorem 3 (Optimal Announcement Policy)**.** *Let* θ_{dyn}^{A*}*,* θ_{pat}^{A*}*, and* θ_{tot}^{A*} *denote optimal announcements under dynastic, paternalistic, and total utilitarian criteria respectively. Suppose the planner's posterior mean is* $\hat{\theta}$ *with precision* τ_ν*.*

(i) ***Dynastic criterion:*** *The optimal announcement is truthful:* $\theta^{A*}_{dyn} = \hat{\theta}$. *The planner respects household preferences and simply provides information.*

(ii) ***Paternalistic criterion:*** *The optimal announcement shades toward the planner's belief about truth:*

$$\theta^{A*}_{pat} = \hat{\theta} + \frac{\tau_A}{\tau_\nu + \tau_A} \cdot \kappa_{pat} \cdot (\hat{\theta} - \bar{\theta}), \tag{40}$$

where $\kappa_{pat} > 0$ *reflects the welfare gain from correcting mean beliefs. When* $\hat{\theta} > \bar{\theta}$ *(planner believes world is more Romerian than households think), the optimal announcement exceeds the truthful benchmark.*

(iii) ***Total utilitarian criterion:*** *The optimal announcement most aggressively promotes high fertility:*

$$\theta^{A*}_{tot} = \hat{\theta} + \frac{\tau_A}{\tau_\nu + \tau_A} \cdot \kappa_{tot} \cdot (\theta_R - \bar{\theta}), \tag{41}$$

where $\kappa_{tot} > \kappa_{pat}$. *In the limit of high credibility, the total utilitarian planner announces* θ_R *regardless of the signal, to maximize population.*

The proof is in Appendix B. The intuition is as follows. The dynastic planner respects preferences as given and gains nothing from distorting information—households know their own preferences, and the planner's role is purely informational. The paternalistic planner wants to correct "mistakes" caused by wrong beliefs, so shades the announcement toward what they believe is true. The total utilitarian planner wants more people regardless of per-capita welfare (as long as $u(c) > 0$), so promotes optimism to raise fertility. Corollary 3 in Appendix B formalizes the resulting ordering: in a Romerian world where the planner's posterior mean is close to θ_R, $\theta^{A*}_{dyn} \leq \theta^{A*}_{pat} \leq \theta^{A*}_{tot}$—the total utilitarian planner most aggressively promotes Romerianism, while the dynastic planner is most conservative.

7.6 The Planner's Dilemma in the Informational Poverty Trap

A key insight emerges from the structure of the learning problem: the planner faces the same informational constraints as households. If the economy is in the uninformative region ($N_t \approx 1$), then the historical productivity signals that inform the planner's estimate $\hat{\theta}$ are also uninformative. The planner's signal precision τ_ν is endogenously low precisely when the trap is most binding.

Remark 3 (The Planner's Dilemma)**.** *Public information is most valuable in the informational poverty trap, where coordinating beliefs could push population above* N^{crit} *and initiate escape. But this is precisely when the planner has the least information: the same uninformative signals that trap households also limit what the planner can learn. The cases where* *intervention matters most are the cases where the planner is most uncertain about what to say.*

This creates a genuine policy dilemma that distinguishes my setting from standard analyses of public information. In Morris and Shin (2002), the concern about public information is that it may be *too influential*, crowding out useful private information. Here, the concern is more subtle: the planner may have *nothing useful to say* in the states where saying something would matter most.

7.7 State-Dependent Value of Announcement

The value of public information depends critically on where the economy sits in the state space. In Appendix B, Proposition 10 shows that when the economy is in the trap region in a Romerian world, the expected value of truthful announcement under paternalistic welfare can be written as the difference between the probability-weighted gain from initiating escape, $P^{esc} \cdot \Delta W^{esc}$, and the probability-weighted loss from deepening the trap, $P^{deepen} \cdot \Delta W^{deepen}$. There exists a precision threshold $\underline{\tau}^{trap}$ above which the expected value is positive. Crucially, because the gain from escape ΔW^{esc} is large (escape avoids extended stagnation), even a moderately noisy signal can be worth announcing.

By contrast, Proposition 11 in Appendix B establishes that in a Malthusian world exhibiting cycles, the value of announcement comes from reducing belief dispersion (and thereby dampening oscillations) but is partially offset by adjustment costs from disrupting existing cyclical dynamics. For sufficiently precise signals the value is positive, but its magnitude is smaller than the trap-escape gain.

7.8 The Stabilizing Role of Malthusians

An additional nuance distinguishes my setting from pure-complementarity coordination games. The heterogeneous strategic interaction—Romerians exhibit complementarity, Malthusians exhibit substitutability—provides a natural stabilizer.

Remark 4 (Malthusians as Stabilizers)**.** *In response to a public announcement, Romerians coordinate (amplifying the signal), but Malthusians anti-coordinate (dampening the signal). This heterogeneity reduces the risk of over-coordination on noise. Compared to a population of only Romerians, a mixed population is safer to inform because Malthusian substitutability absorbs some of the coordination externality.*

This implies that the precision threshold for beneficial announcement is *lower* in populations with substantial Malthusian minorities than in purely Romerian populations. Paradoxically, the presence of pessimists makes communication policy safer.

7.9 Comparison of Decentralized and Planner Allocations

Table 6 summarizes how decentralized allocations compare to the planner's optimum under each welfare criterion.

Table 6. Decentralized vs. Planner Allocations by Welfare Criterion

	Dynastic	Paternalistic	Total Utilitarian
Romerian World ($\theta^* > 0$)			
Decentralized fertility	Too low (externality + wrong beliefs)		
Optimal announcement	Truthful	Shade toward θ_R	Strongly toward θ_R
Residual inefficiency	Externality only	Externality only	Externality only
Malthusian World ($\theta^* < 0$)			
Decentralized fertility	Too high (externality), mixed by beliefs		
Optimal announcement	Truthful	Shade toward θ_M	Ambiguous
Residual inefficiency	Externality only	Externality only	Externality + pop. loss

Under all criteria, the decentralized equilibrium is inefficient due to the uninternalized scale-effect externality. Optimal announcement policy can address belief distortions but not the externality itself. Additional instruments (fertility subsidies in Romerian worlds, fertility taxes in Malthusian worlds) would be needed to achieve full efficiency.

7.10 Policy Implications

The analysis yields several implications for government communication policy.

In the trap, take calculated risks. When the economy appears trapped (low fertility, stagnant population, persistent pessimism), even a moderately imprecise signal may be worth announcing if it has a chance of coordinating beliefs and initiating escape. The expected value of escape is large enough to justify risk-taking under paternalistic or total utilitarian criteria.

Under dynastic welfare, be truthful. If the planner respects household preferences as given, the optimal policy is simply to provide accurate information. Distorting announcements would make some households worse off according to their own preferences.

During Malthusian cycles, time announcements carefully. In a Malthusian world, crises naturally correct beliefs. Announcing during an expansion (when population is high and signals are informative) can accelerate learning and dampen cycles. Announcing during a crisis is less valuable because beliefs are already moving toward Malthusianism.

Build credibility for when it matters. The value of announcement depends on credibility (τ_A). Planners should invest in credibility during normal times (by making accurate predictions) so that announcements are effective when intervention matters most—in the trap.

Recognize the limits of knowledge. The most important policy implication is epistemic humility. When the economy is in the trap, the planner knows the least about what to say. Acknowledging this uncertainty, rather than projecting false confidence, may be the most honest policy.

8 Conclusion

Beliefs about the economic consequences of population are not merely background assumptions in fertility models. They are first-order determinants of demographic and macroeconomic dynamics. This paper has formalized that observation in an overlapping generations model with heterogeneous beliefs about scale effects, and has shown that the resulting feedback between beliefs, fertility, and the informativeness of data produces a sharp duality. Malthusian minorities trap Romerian economies in stagnation, while Romerian minorities destabilize Malthusian economies into cycles. Wrong beliefs are always costly, but the form of the cost is determined by the truth they fail to track.

The informational poverty trap is the paper's most distinctive theoretical object. Unlike traps grounded in technological nonconvexities or credit frictions, it arises endogenously from the geometry of Bayesian learning. Signals are uninformative precisely in the region where pessimism keeps the economy confined, so the very beliefs that cause stagnation also insulate themselves from correction. This mechanism reframes a long-standing question in economic history. The pre-industrial Malthusian era can be read as a true Malthusian regime that ended with a technological break, or as a Romerian regime trapped for millennia by self-reinforcing pessimism that finally dissolved when population breached the informative region. The two readings are observationally similar in qualitative terms but carry sharply different implications for whether the modern fertility decline represents a benign demographic adjustment or a slow drift back into the trap.

Belief heterogeneity also forces a reckoning with how welfare itself is defined. Because the population over which welfare is evaluated is endogenous to the policy being evaluated, dynastic, paternalistic, and utilitarian criteria recommend systematically different communication strategies, and the gap between them widens precisely in the regions where intervention matters most. The same logic produces an unconventional message about public information. Rather than the standard warning that public signals can over-coordinate beliefs, the binding constraint here is that the planner is most uncertain exactly when announcements would be most valuable. The heterogeneous strategic interaction, with complementarity among Romerians and substitutability among Malthusians, softens but does not resolve this tension.

Several limitations point toward natural extensions. The empirical results establish a robust conditional correlation between Malthusian beliefs and fertility but stop short of causal identification. Instrumental variation in belief exposure, perhaps through plausibly exogenous shifts in environmental discourse or schooling content, would sharpen the case. The model also abstracts from the quantity-quality margin central to unified growth theory, and embedding human capital choice would let the framework speak directly to the demographic transition rather than treating it as an interpretive lens. Open-economy extensions with migration would introduce a new channel through which heterogeneous priors shape aggregate dynamics, namely belief-driven sorting across borders. Finally, a quantitative implementation disciplined by long-run population and productivity data could in principle adjudicate between the two interpretations of pre-industrial history that the theory leaves open.

These extensions are worth pursuing because the framework they would build on does something the existing literature does not. Models of endogenous fertility typically take the population-productivity relationship as known. Models of heterogeneous beliefs typically take the set of agents as fixed. This paper joins the two, and in doing so it identifies a class of dynamics in which the population that generates the data and the beliefs that interpret the data co-evolve. The informational poverty trap, the Romerian-Malthusian duality, and the state-dependent value of public information all follow from taking that joint evolution seriously. Whether one reads human history as a Malthusian regime that ended or a Romerian regime that nearly never began, the lesson is the same. What societies believe about themselves shapes what they become, and what they become determines what they can learn.

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A Proofs and Auxiliary Theoretical Results

This appendix contains proofs of the main-text results, together with auxiliary theoretical results referenced from the main text.

A.1 Proof of Lemma 1

I solve the household optimization problem and verify the value function conjecture.

Step 1: Bellman equation. The Bellman equation for type j is:

$$V_j(A, N, \phi) = \max_{n \geq 0} \left\{ \ln(w - \psi(n)) + \beta n \cdot \mathbb{E}_j[V_j(A', N', \phi')] \right\}, \tag{42}$$

where $w = \alpha A N^{\alpha-1}$ and expectations are taken under belief θ_j. Because households are atomistic in a continuum, individual fertility n does not affect the aggregate state (A', N', ϕ'); the choice variable enters only through the flow utility $\ln(w - \psi(n))$ and the multiplier βn in front of the continuation value.

Step 2: Conjecture. Conjecture $V_j(A, N, \phi) = v_j(\phi) + \xi_{A,j} \ln A + \xi_{N,j} \ln N$, where the constants $\xi_{A,j}, \xi_{N,j}$ are allowed to depend on type and on the equilibrium fertility n_j^*. The flow utility $\ln(w - \psi(n)) = \ln \alpha + \ln A + (\alpha - 1) \ln N + \ln(1 - \psi(n)/w)$ admits a log-linear part plus a residual term $\ln(1 - \psi(n)/w)$ that is bounded for interior n and is absorbed into $v_j(\phi)$ at leading order in the log-linear approximation.

Step 3: Expected continuation value. Under the laws of motion (taking the log-linear approximation $\ln(1 + x) \approx x$ for small x):

$$\ln A' \approx \ln A + \bar{g} + \theta_j \ln N, \tag{43}$$

$$\ln N' = \ln N + \ln \bar{n}. \tag{44}$$

Thus

$$\begin{aligned} \mathbb{E}_j[\xi_{A,j} \ln A' + \xi_{N,j} \ln N'] = {} & \xi_{A,j} \ln A + \xi_{N,j} \ln N \\ & + \xi_{A,j} \bar{g} + \xi_{A,j} \theta_j \ln N + \xi_{N,j} \ln \bar{n}. \end{aligned} \tag{45}$$

Step 4: Matching coefficients. At the optimal n_j^*, substituting the conjecture into the Bellman equation and matching coefficients of $\ln A$ and $\ln N$ on both sides gives:

$$\xi_{A,j} = 1 + \beta n_j^* \xi_{A,j} \quad \Rightarrow \quad \xi_{A,j} = \frac{1}{1 - \beta n_j^*}, \tag{46}$$

$$\xi_{N,j} = (\alpha - 1) + \beta n_j^* \big(\xi_{A,j}\theta_j + \xi_{N,j}\big) \quad \Rightarrow \quad \xi_{N,j} = \frac{(\alpha - 1)(1 - \beta n_j^*) + \beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}. \tag{47}$$

Both coefficients depend on the type-specific equilibrium fertility n_j^* and (in the case of $\xi_{N,j}$) explicitly on θ_j. The standing assumption $\beta n_j^* < 1$ ensures finiteness.

Step 5: First-order condition. With the quadratic cost $\psi(n) = \frac{\gamma}{2}n^2$, the FOC is:

$$\frac{\gamma n}{w - \frac{\gamma}{2}n^2} = \beta \mathbb{E}_j[V_j(A', N', \phi')]. \tag{48}$$

Holding $(A, N, \bar{n})$ at their equilibrium values, the right-hand side is strictly increasing in θ_j: the term $\xi_{A,j}\theta_j \ln N$ is increasing in θ_j for $\ln N > 0$, and $\xi_{N,j}$ is itself increasing in θ_j for fixed n_j^* with $\beta n_j^* < 1$. Together with the implicit-function argument over the system that simultaneously pins down n_j^* and the coefficients, the resulting equilibrium fertility n_j^* is strictly increasing in θ_j. We can therefore write $n_j^* = n^* \cdot h(\theta_j; \bar{n})$ for a baseline level n^* and a function h that is increasing in its first argument.

Step 6: Fertility ranking. Since h is increasing and $\theta_R > \theta_M$, we have $n_R^* > n_M^*$. □

A.2 Proof of Proposition 1

I derive the dependence of optimal fertility on expected aggregate fertility.

Step 1: Setup. Consider type j's problem when the household expects aggregate fertility $\bar{n}^e$. The continuation value is

$$\mathbb{E}_j[V_j(A', N', \phi')] = v_j(\phi') + \xi_{A,j}[\ln A + \bar{g} + \theta_j \ln N] + \xi_{N,j}[\ln N + \ln \bar{n}^e], \tag{49}$$

where $\xi_{A,j}$ and $\xi_{N,j}$ are as derived in the proof of Lemma 1. Crucially, current-period TFP growth depends on current population N, not on $\bar{n}^e$; the dominant direct dependence of the continuation value on $\bar{n}^e$ comes through $N' = \bar{n}^e N$, and is captured by the coefficient $\xi_{N,j}$ (which encapsulates the discounted infinite stream of future scale-effect and dilution effects). The indirect channel through ϕ' enters $v_j(\phi')$ and is of higher order for the local sign analysis.

Step 2: Sign of the marginal continuation value. Differentiating,

$$\frac{\partial \mathbb{E}_j[V_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}, \tag{50}$$

and from Step 4 of the previous proof, the sign of $\xi_{N,j}$ equals the sign of

$$(\alpha - 1)(1 - \beta n_j^*) + \beta n_j^* \theta_j. \tag{51}$$

For Malthusians ($\theta_M < 0$): both $(\alpha - 1)(1 - \beta n_M^*) < 0$ and $\beta n_M^* \theta_M < 0$, so $\xi_{N,M} < 0$ unambiguously.

For Romerians ($\theta_R > 0$): the second term is positive but the first is negative; $\xi_{N,R} > 0$ iff $\theta_R > (1 - \alpha)(1 - \beta n_R^*)/(\beta n_R^*)$. This is condition (22), which is assumed.

Step 3: Effect on optimal fertility. From the FOC,

$$\frac{\gamma n_j}{w - \frac{\gamma}{2} n_j^2} = \beta \mathbb{E}_j[V_j], \tag{52}$$

the left-hand side is strictly increasing in n_j, so $\text{sign}(dn_j^*/d\bar{n}^e) = \text{sign}(\partial \mathbb{E}_j[V_j]/\partial \bar{n}^e) = \text{sign}(\xi_{N,j})$.

Combining with Step 2: $dn_R^*/d\bar{n}^e > 0$ (strategic complementarity for Romerians, under (22)) and $dn_M^*/d\bar{n}^e < 0$ (strategic substitutability for Malthusians). $\square$

A.3 Slope of the Aggregate Best Response (Auxiliary Result)

Corollary 1 (Slope of Aggregate Best Response)**.** *Under the conditions of Proposition 1, the aggregate best response function* $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi) n_M^*(\bar{n}^e)$ *has slope:*

$$\frac{dBR}{d\bar{n}^e} = \phi \frac{dn_R^*}{d\bar{n}^e} + (1 - \phi) \frac{dn_M^*}{d\bar{n}^e}. \tag{53}$$

This slope is:

(i) Positive when ϕ is sufficiently large (Romerian dominance);

(ii) Negative when ϕ is sufficiently small (Malthusian dominance);

(iii) Zero at a critical ϕ^* where the two effects exactly offset.

Proof. The slope formula follows by differentiation. From Proposition 1, $dn_R^*/d\bar{n}^e > 0$ and $dn_M^*/d\bar{n}^e < 0$. Therefore the slope is positive (resp. negative) for ϕ close to 1 (resp. 0), and continuity in ϕ implies the existence of a unique zero ϕ^*, given explicitly by (23). $\square$

A.4 Proof of Proposition 2

Step 1: Aggregate best response. Define $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1-\phi)n_M^*(\bar{n}^e)$.

Step 2: Slope. Use Corollary 1.

Step 3: Uniqueness condition. An equilibrium requires $\bar{n}^e = BR(\bar{n}^e)$. Graphically, this is the intersection of the 45-degree line with $BR(\bar{n}^e)$.

When $\phi < \phi^*$, BR slopes downward. Since $BR(0) > 0$ and $BR(\bar{n}^e)$ is bounded, there is exactly one intersection: unique equilibrium.

When $\phi > \phi^*$, BR slopes upward. If the slope exceeds 1 at some point, the curves can intersect multiple times, generating multiple equilibria.

Step 4: Ranking equilibria. When multiple equilibria exist, they can be ordered. Let $\bar{n}^L < \bar{n}^M < \bar{n}^H$ be three equilibria (if they exist). Standard arguments show that $\bar{n}^L$ and $\bar{n}^H$ are stable under best-response dynamics while $\bar{n}^M$ is unstable. □

A.5 Equilibrium Selection under Incomplete Information (Auxiliary Result)

Proposition 6 (Equilibrium Selection)**.** *Consider the parameter region in which Proposition 2(ii) holds, so that the complete-information game admits a low-fertility and a high-fertility equilibrium. As signal noise $\sigma_\eta \to 0$, there exists a unique equilibrium of the incomplete-information game in which each type uses a threshold strategy on its private signal:*

(i) Romerians coordinate on the high-fertility equilibrium n_R^H when $\phi_i > \phi^{**}$ and on the low-fertility equilibrium n_R^L when $\phi_i < \phi^{**}$;

(ii) Malthusians choose their (substitute) low-fertility response n_M^L when $\phi_i > \phi^{**}$ and their high-fertility response n_M^H when $\phi_i < \phi^{**}$;

where the common threshold ϕ^{**} depends on model parameters.

Proof. The proof adapts the global games equilibrium selection argument of Morris and Shin (1998) to our setting with heterogeneous strategic interaction.

Step 1: Setup with private signals. Each household observes a private signal $\phi_i = \phi + \eta_i$ where $\eta_i \sim N(0, \sigma_\eta^2)$ i.i.d. The true ϕ is drawn from a prior; for simplicity, assume $\phi \sim U[0,1]$. We restrict attention to the parameter region of Proposition 2(ii), in which the complete-information game admits two stable equilibria, naturally labeled "low" and "high" by aggregate fertility.

Step 2: Threshold strategies. Consider symmetric threshold strategies in which each type's equilibrium choice in the continuous-action game is selected by comparing the private signal to a threshold: Romerians choose n_R^H if $\phi_i > \phi^{**}$, n_R^L if $\phi_i < \phi^{**}$; Malthusians choose n_M^L if $\phi_i > \phi^{**}$, n_M^H if $\phi_i < \phi^{**}$. The Malthusian strategy is reversed because of strategic substitutability: when Malthusians infer that the Romerian share is high (and hence aggregate fertility is high), they reduce their own fertility.

Step 3: Indifference condition. At the threshold ϕ^{**}, each type must be indifferent between the high and low equilibrium choices. This requires that the expected aggregate fertility implied by signal ϕ^{**} makes both types simultaneously indifferent at ϕ^{**}.

For a household of type j with signal $\phi_i = \phi^{**}$, the expected aggregate fertility integrates over the posterior on ϕ and over the resulting fractions of agents above and below the threshold. As $\sigma_\eta \to 0$, ϕ_i becomes increasingly informative about ϕ, and the conditional distribution of the rank of ϕ^{**} within the population converges to the uniform Laplacian distribution on $[0, 1]$, the standard global-games limit.

Step 4: Fixed point. The threshold ϕ^{**} is determined by a fixed-point condition combining the Romerian and Malthusian indifference equations. Because the Romerian indifference equation is monotone in one direction and the Malthusian indifference equation is monotone in the other direction (a consequence of opposite strategic interactions), the fixed point is unique provided both indifference functions are continuous and intersect—which they do under standard regularity.

Step 5: Uniqueness. By the standard global games iterated dominance argument applied to the binary equilibrium-selection game, as $\sigma_\eta \to 0$ the unique threshold equilibrium is selected. This pins down a unique outcome from among the multiple equilibria that exist under common knowledge. $\square$

A.6 Proof of Theorem 1

I show convergence in Romerian worlds using a Bayesian-consistency argument combined with a deterministic monotonicity argument for demographic selection.

Step 1: Dynamics. The Romerian share evolves as:

$$\phi_{t+1} = \lambda \phi_t^{birth} + (1 - \lambda)\pi_{t+1}, \tag{54}$$

where $\phi_t^{birth} = \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t) n_M}$ and π_{t+1} is the Bayesian posterior.

Step 2: Demographic selection. For $\phi_t \in (0,1)$, since $n_R > n_M$:

$$\phi_t^{birth} > \phi_t \iff \frac{\phi_t n_R}{\phi_t n_R + (1-\phi_t)n_M} > \phi_t \iff n_R > \phi_t n_R + (1-\phi_t)n_M, \tag{55}$$

and the last inequality reduces to $(1-\phi_t)(n_R - n_M) > 0$, which holds. Thus $\phi_t^{birth} > \phi_t$ strictly for any interior ϕ_t.

Step 3: Learning. In a Romerian world ($\theta^* = \theta_R$), the signal is $s_t = \bar{g} + \theta_R \ln N_t + \varepsilon_t$. A direct computation yields

$$\ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)} = \frac{(\theta_R - \theta_M)\ln N_t \cdot \left[2(s_t - \bar{g}) - (\theta_R + \theta_M)\ln N_t\right]}{2\sigma_\varepsilon^2}, \tag{56}$$

and substituting $s_t - \bar{g} = \theta_R \ln N_t + \varepsilon_t$ and taking expectations under the true Romerian model gives

$$\mathbb{E}^{\theta_R}\left[\ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)} \,\middle|\, \mathcal{F}_{t-1}\right] = \frac{(\theta_R - \theta_M)^2(\ln N_t)^2}{2\sigma_\varepsilon^2} \geq 0 \tag{57}$$

with equality if and only if $\ln N_t = 0$. Standard Bayesian-consistency results then imply that under the true Romerian measure, $\{\pi_t\}$ is a bounded $[0,1]$-valued submartingale, and by the submartingale convergence theorem π_t converges almost surely to a limit $\pi_\infty \in [0,1]$.

To show $\pi_\infty = 1$ a.s., note that the cumulative information $\sum_{t=0}^{\infty} \mathcal{I}_t$ diverges a.s. whenever $(\ln N_t)^2 \to \infty$. Under sustained divergence of $|\ln N_t|$, the log-likelihood ratio diverges and $\pi_t \to 1$. If, by way of contradiction, population remained bounded forever, then $\bar{n}_t$ would have to stay weakly below 1 persistently, which would require $\phi_t \leq \phi^{trap}$ persistently. But demographic selection deterministically pushes $\phi_t^{birth} > \phi_t$ at each step (Step 2); combined with the bounded contribution from learning, ϕ_t cannot remain forever bounded below ϕ^{trap}. So eventually $\bar{n}_t > 1$ and population grows unboundedly, forcing $\pi_t \to 1$ a.s.

Step 4: Combined dynamics. For any interior (ϕ_t, π_t), $\phi_t^{birth} > \phi_t$ deterministically, and $\pi_t \to 1$ a.s. by Step 3. From (14), and using $1 - \phi_t^{birth} < 1 - \phi_t$,

$$1 - \phi_{t+1} = \lambda(1 - \phi_t^{birth}) + (1-\lambda)(1 - \pi_{t+1}) \leq \lambda(1 - \phi_t) + (1-\lambda)(1 - \pi_{t+1}).$$

Fix any $\varepsilon > 0$. Almost surely, there exists $T(\omega)$ such that for all $t \geq T$, $1 - \pi_{t+1} < \varepsilon$. Iterating the inequality from T gives

$$1 - \phi_{T+k} \leq \lambda^k(1 - \phi_T) + (1-\lambda)\varepsilon \cdot \sum_{j=0}^{k-1} \lambda^j = \lambda^k(1 - \phi_T) + \varepsilon(1 - \lambda^k).$$

Hence $\limsup_{k\to\infty}(1-\phi_{T+k}) \leq \varepsilon$ a.s. Since ε is arbitrary, $\phi_t \to 1$ almost surely.

Step 5: Conclusion. Aggregate fertility $\bar{n}_t = \phi_t n_R + (1-\phi_t)n_M$ converges to n_R^*, and population grows at rate $n_R^* - 1$ in the long run. □

A.7 Proof of Proposition 3

Step 1: Learning bound. In the uninformative region $|\ln N_t| < \delta$, the per-period expected log-likelihood ratio is bounded by:

$$\mathbb{E}\left[\ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)} \,\middle|\, \mathcal{F}_t\right] \leq \frac{(\theta_R - \theta_M)^2 \delta^2}{2\sigma_\varepsilon^2} = \bar{\mathcal{I}}(\delta). \tag{58}$$

Local linearization of the Bayes update around an interior prior gives

$$\pi_{t+1} - \pi_t \approx \pi_t(1-\pi_t) \cdot \ln \frac{\ell(s_t|\theta_R)}{\ell(s_t|\theta_M)}, \tag{59}$$

so the expected change in the Romerian posterior per period (driving the learning channel for children who do not inherit) is bounded by:

$$\mathbb{E}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] \leq C_1 \cdot \bar{\mathcal{I}}(\delta), \tag{60}$$

where $C_1 = \max_{\pi\in[0,1]} \pi(1-\pi) = 1/4$. This establishes (28).

Step 2: Selection bound. The change in Romerian share from demographic selection is:

$$\phi_t^{birth} - \phi_t = \frac{\phi_t(1-\phi_t)(n_R - n_M)}{\phi_t n_R + (1-\phi_t)n_M} \leq \frac{(n_R - n_M)(1-\phi_t)}{n_M}, \tag{61}$$

since $\bar{n}_t \geq n_M > 0$ and $\phi_t \leq 1$. Setting $C_2 = 1/n_M$ gives (29).

Step 3: Escape time bound. Combining the two channels and using that, in the trap region under sustained pessimism, the expected per-period drift in ϕ_t is dominated by the sum of the demographic-selection bound and the learning bound:

$$\mathbb{E}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] \leq \lambda C_2(n_R - n_M)(1-\phi_t) + (1-\lambda)C_1\bar{\mathcal{I}}(\delta) \leq \lambda C_2(n_R - n_M) + (1-\lambda)C_1\bar{\mathcal{I}}(\delta). \tag{62}$$

By definition of T^{esc}, at the stopping time $\phi_{T^{esc}} \geq \bar{\phi}$, so $\mathbb{E}[\phi_{T^{esc}} - \phi_0] \geq \bar{\phi} - \phi_0$. Combined with the optional-stopping bound $\mathbb{E}[\phi_{T^{esc}} - \phi_0] \leq \mathbb{E}[T^{esc}] \cdot D$, where $D \equiv \lambda C_2(n_R - n_M) +$

$(1-\lambda)C_1\bar{\mathcal{I}}(\delta)$ is the per-period drift bound, this gives

$$\bar{\phi} - \phi_0 \;\leq\; \mathbb{E}[\phi_{T^{esc}} - \phi_0] \;\leq\; \mathbb{E}[T^{esc}] \cdot D, \tag{63}$$

so that

$$\mathbb{E}[T^{esc}] \;\geq\; \frac{\bar{\phi} - \phi_0}{\lambda C_2(n_R - n_M) + (1-\lambda)C_1\bar{\mathcal{I}}(\delta)}. \tag{64}$$

This establishes (30). □

A.8 Trap Duration (Auxiliary Result)

Corollary 2 (Trap Duration)**.** *Under the conditions of Proposition 3, the expected escape time diverges as:*

$$\mathbb{E}[T^{esc}] = \Omega\left(\frac{1}{\delta^2}\right) \quad \textit{as } \delta \to 0, \tag{65}$$

holding other parameters fixed. For sufficiently small δ, the expected escape time can exceed any finite horizon.

Proof. By definition, $\bar{\mathcal{I}}(\delta) = (\theta_R - \theta_M)^2\delta^2/(2\sigma_\varepsilon^2) = O(\delta^2)$. The lower bound in Proposition 3 can therefore be written as

$$\mathbb{E}[T^{esc}] \;\geq\; \frac{\bar{\phi} - \phi_0}{a + b\delta^2},$$

where $a = \lambda C_2(n_R - n_M) \geq 0$ and $b = (1-\lambda)C_1(\theta_R - \theta_M)^2/(2\sigma_\varepsilon^2) > 0$. When $a = 0$ (no cultural transmission, or $n_R = n_M$), this gives $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/(b\delta^2) = \Omega(1/\delta^2)$ directly. When $a > 0$, the bound remains finite for any $\delta > 0$, but in the regime where demographic selection is weak relative to learning—which is the relevant case for diagnosing the trap—one has $a \ll b\delta^2$ only outside the trap; *within* the trap region the relevant comparison is between escape times at different values of δ, and the δ^2 scaling controls how quickly the bound shrinks as δ grows. The qualitative implication—that the lower bound grows without bound as $\delta \to 0$ when learning is the dominant escape channel—establishes the corollary. □

A.9 Trap Boundary (Auxiliary Result)

Proposition 7 (Trap Boundary)**.** *Define the* critical population $N^{crit}(\phi)$ *as the minimum population such that learning dominates demographic selection in driving belief dynamics:*

$$N^{crit}(\phi) = \exp\left(\sqrt{\frac{2\sigma_\varepsilon^2 \cdot \lambda C_2(n_R - n_M)(1-\phi)}{(1-\lambda)C_1(\theta_R - \theta_M)^2}}\right). \tag{66}$$

Then:

(i) For $N_t < N^{crit}(\phi_t)$, demographic selection dominates learning, and ϕ_t increases slowly.

(ii) For $N_t > N^{crit}(\phi_t)$, learning dominates demographic selection, and ϕ_t increases rapidly toward 1 (in a Romerian world).

(iii) The economy escapes the trap when population crosses N^{crit}; before that, it is effectively trapped.

Proof.

Step 1: Define the boundary. The critical population $N^{crit}(\phi)$ is defined by equating the learning and demographic selection contributions to ϕ dynamics:

$$(1-\lambda)C_1 \cdot \frac{(\theta_R - \theta_M)^2(\ln N^{crit})^2}{2\sigma_\varepsilon^2} = \lambda C_2(n_R - n_M)(1-\phi). \tag{67}$$

Solving for N^{crit}:

$$(\ln N^{crit})^2 = \frac{2\sigma_\varepsilon^2 \lambda C_2(n_R - n_M)(1-\phi)}{(1-\lambda)C_1(\theta_R - \theta_M)^2}. \tag{68}$$

Taking the positive root (since $N^{crit} > 1$ for the relevant case):

$$N^{crit}(\phi) = \exp\left(\sqrt{\frac{2\sigma_\varepsilon^2 \lambda C_2(n_R - n_M)(1-\phi)}{(1-\lambda)C_1(\theta_R - \theta_M)^2}}\right). \tag{69}$$

Step 2: Interpretation. For $N_t < N^{crit}(\phi_t)$, the learning contribution is smaller than the demographic selection contribution, so selection dominates. For $N_t > N^{crit}(\phi_t)$, learning dominates. In a Romerian world, crossing N^{crit} initiates rapid convergence toward $\phi = 1$. ☐

A.10 Proof of Theorem 2

I establish the existence of cycles using local bifurcation analysis around an interior steady state.

Step 1: Linearized dynamics. Consider the system in $(\phi_t, \ln N_t)$. Let $\hat{\phi}_t = \phi_t - \phi^{ss}$ and $\hat{n}_t = \ln N_t - \ln N^{ss}$ denote deviations from an interior steady state of the deterministic skeleton.

The linearized dynamics are:

$$\begin{pmatrix} \hat{\phi}_{t+1} \\ \hat{n}_{t+1} \end{pmatrix} = J \begin{pmatrix} \hat{\phi}_t \\ \hat{n}_t \end{pmatrix}, \tag{70}$$

where J is the Jacobian matrix evaluated at the steady state.

Step 2: Jacobian entries. The entries of J are: $J_{11} = \partial\phi_{t+1}/\partial\phi_t > 0$ (more Romerians today mean more Romerian children through demographic selection); $J_{12} = \partial\phi_{t+1}/\partial \ln N_t < 0$ in a Malthusian world (higher N_t leads to lower realized productivity growth, which updates beliefs toward Malthusianism); $J_{21} = \partial \ln N_{t+1}/\partial\phi_t > 0$ (more Romerians mean higher aggregate fertility); $J_{22} = \partial \ln N_{t+1}/\partial \ln N_t = 1 + \partial \ln \bar{n}_t/\partial \ln N_t$, where the sign of the second term depends on how fertility responds to wages and is generally ambiguous but bounded.

Step 3: Eigenvalue analysis. The characteristic polynomial is

$$\mu^2 - \tau\mu + \Delta = 0, \tag{71}$$

with trace $\tau = J_{11} + J_{22}$ and determinant $\Delta = J_{11}J_{22} - J_{12}J_{21}$. Eigenvalues are complex when $\tau^2 < 4\Delta$; when complex, their modulus equals $\sqrt{\Delta}$, so the modulus exceeds one iff $\Delta > 1$.

Step 4: Conditions for oscillatory instability. Since $J_{12} < 0$ and $J_{21} > 0$ in a Malthusian world, the cross-term $-J_{12}J_{21} > 0$ raises Δ. For sufficiently strong learning ($|J_{12}|$ large enough that $\Delta > 1$) but not too strong ($\tau^2 < 4\Delta$), we obtain complex eigenvalues with $|\mu| > 1$.

Step 5: Limit cycle. When learning is very weak ($|J_{12}|$ small), $\Delta < 1$ and eigenvalues are real with modulus ≤ 1; the interior steady state is locally stable and the system converges monotonically. When learning is very strong ($|J_{12}|$ very large), the interior steady state ceases to exist or shifts toward a corner with Malthusian dominance, where Malthusian beliefs prevail; the relevant dynamics are now around that corner, and oscillations cease. For intermediate $|J_{12}|$, the linearized system around the interior steady state is locally explosive with complex eigenvalues. The global nonlinear system is bounded (since $\phi \in [0, 1]$ and population is bounded by carrying capacity), so the explosive linear dynamics translate into bounded oscillations consistent with a stable limit cycle. The standard Neimark–Sacker bifurcation argument then establishes the existence of an attracting invariant closed curve when $|\mu|$ crosses one transversally. □

A.11 Cycle Existence Conditions (Auxiliary Result)

Proposition 8 (Cycle Existence Conditions)**.** *Define $\mathcal{L} = \sigma_\varepsilon^{-2}|\theta_R - \theta_M|$ as an index of learning responsiveness and $\mathcal{D} = \lambda(n_R - n_M)/\bar{n}$ as an index of demographic selection strength.*

The linearized economy exhibits oscillatory instability around an interior steady state (the precondition for the limit cycle described in Theorem 2) if and only if:

$$\underline{\mathcal{L}}(\mathcal{D}) < \mathcal{L} < \overline{\mathcal{L}}(\mathcal{D}), \tag{72}$$

where the threshold functions $\underline{\mathcal{L}}$ *and* $\overline{\mathcal{L}}$ *are characterized below.*

Proof.

Step 1: Define indices. Let $\mathcal{L} = \sigma_\varepsilon^{-2}|\theta_R - \theta_M|$ index learning responsiveness and $\mathcal{D} = \lambda(n_R - n_M)/\bar{n}$ index demographic selection.

Step 2: Thresholds. From the eigenvalue analysis in the proof of Theorem 2: $\underline{\mathcal{L}}(\mathcal{D})$ is the smallest $\mathcal{L}$ such that $\Delta > 1$ holds; below this, demographic selection dominates and $\phi \to 1$ monotonically; $\overline{\mathcal{L}}(\mathcal{D})$ is the largest $\mathcal{L}$ consistent with the existence of an interior steady state and $\tau^2 < 4\Delta$; above this, the relevant fixed point migrates to a corner and oscillation around the interior point ceases.

Step 3: Derivation. Setting $\Delta = 1$ and $\tau^2 = 4\Delta$ yields two equations in the entries of J, which are themselves functions of $\mathcal{L}$ and $\mathcal{D}$. Solving for $\mathcal{L}$ at each boundary gives the threshold functions. The algebra is standard but tedious and is omitted. □

B Welfare Analysis Proofs and Auxiliary Results

B.1 Proof of Proposition 4

Step 1: Planner's problem. The planner maximizes

$$\sum_{t=0}^{\infty} \beta^t N_t \cdot u(c_t - \psi(n_t)/N_t) \tag{73}$$

where I write the cost as per-capita for clarity. With $c_t = A_t N_t^{\alpha-1}$, the planner's first-order condition for n_t is:

$$\psi'(n_t) = \beta \frac{\partial}{\partial n_t} \mathbb{E}\left[\sum_{\tau>t} \beta^{\tau-t-1} N_\tau u(c_\tau)\right]. \tag{74}$$

Step 2: Externality. The key difference from the household problem is that the planner accounts for $\partial N_\tau / \partial n_t = N_t \cdot \prod_{s=t+1}^{\tau-1} n_s$ for $\tau > t$, and for $\partial A_\tau/\partial n_t$ through the chain of TFP equations. In a Romerian world, $\partial A_\tau / \partial N_{t+1} > 0$, so the planner perceives a higher marginal benefit of fertility than households, implying $n^{FB} > n^{CB}$.

Step 3: Belief distortion. Since Malthusians believe $\theta_M < 0 < \theta^*$, they perceive a lower (indeed negative) scale effect and choose $n_M^* < n^{CB} < n^{FB}$. □

B.2 First-Best under Alternative Welfare Criteria (Auxiliary Result)

Proposition 9 (First-Best under Alternative Criteria). *Let $n^{FB,pat}$, $n^{FB,tot}$, and $n^{FB,avg}$ denote first-best fertility under paternalistic, total utilitarian, and average utilitarian criteria respectively. In a Romerian world:*

$$n^{FB,avg} < n^{FB,pat} < n^{FB,tot}. \tag{75}$$

The total utilitarian planner chooses the highest fertility; the average utilitarian planner is indifferent to population size and chooses fertility based only on the per-capita tradeoff.

Proof. Under total utilitarianism, the planner's objective includes an additional term $N_t u(c_t)$ that values having more people directly. This pushes optimal fertility above the paternalistic level. Under average utilitarianism, the planner is indifferent to N_t per se, caring only about $u(c_t)$. The optimal fertility then balances the cost $\psi(n)$ against the future per-capita consumption benefit, without the population-size bonus of total utilitarianism or the internalized externality effect of paternalistic welfare. □

B.3 Proof of Proposition 5

Step 1: Define benchmarks. Let $W^{FB}(\theta^*)$ be first-best welfare, $W^{CB}(\theta)$ be common-belief decentralized welfare when all households believe θ, and W^{HB} be heterogeneous-belief welfare.

Step 2: Decomposition.

$$\begin{aligned} W^{FB} - W^{HB} &= (W^{FB} - W^{CB}(\theta^*)) + (W^{CB}(\theta^*) - W^{CB}(\bar{\theta})) + (W^{CB}(\bar{\theta}) - W^{HB}) && (76) \\ &= \Delta_1 + \Delta_2 + \Delta_3. && (77) \end{aligned}$$

Step 3: Quadratic approximation for Δ_3. See the lemma proved next. □

B.4 Lemma: Welfare Loss from Belief Dispersion

Lemma. *Aggregate (true-evaluation) welfare under heterogeneous beliefs satisfies*

$$W^{CB}(\theta^*) - W^{HB} \approx \frac{\omega\kappa^2}{2}\big[\,Var(\theta_i) + (\bar{\theta} - \theta^*)^2\big],$$

where $\omega = -V''(n^{CB}) > 0$ is the curvature of the true-evaluation value function at the common-belief optimum and κ is the slope of the fertility policy in beliefs. When $\bar{\theta} = \theta^*$, this reduces to $\Omega \cdot Var(\theta_i)$ with $\Omega = \omega\kappa^2/2 > 0$.

Proof.

Step 1: Setup. Aggregate welfare under heterogeneous beliefs is

$$W^{HB} = \phi V_R + (1-\phi)V_M, \tag{78}$$

where V_j is the (true) value attained by a type-j household choosing fertility n_j^* under belief θ_j, evaluated under the true law of motion. The benchmark welfare $W(\theta^*)$ is the corresponding object in the counterfactual common-belief decentralized equilibrium where every household holds the correct belief θ^* and chooses the resulting individually-optimal fertility n^{CB}.

Step 2: Quadratic approximation. Let n^{CB} denote the common-belief equilibrium fertility. By the household's first-order condition at the correct-belief optimum, the (true-evaluation) value function $V(n)$ viewed as a function of own fertility n satisfies $V'(n^{CB}) = 0$. Standard Taylor expansion then gives

$$V^* - V_j \approx \frac{\omega}{2}(n_j^* - n^{CB})^2, \tag{79}$$

where $V^* = V(n^{CB})$ and $\omega \equiv -V''(n^{CB}) > 0$ is the curvature of the (true-evaluation) value at n^{CB}.

Step 3: Linking fertility to beliefs. Linearizing the policy function around θ^*, $n_j^* - n^{CB} \approx \kappa(\theta_j - \theta^*) + o(\theta_j - \theta^*)$, where $\kappa > 0$ is the slope of the fertility policy in beliefs.

Step 4: Aggregate loss. Combining the previous two steps,

$$W(\theta^*) - W^{HB} = \phi(V^* - V_R) + (1-\phi)(V^* - V_M)$$
$$\approx \frac{\omega\kappa^2}{2}\left[\phi(\theta_R - \theta^*)^2 + (1-\phi)(\theta_M - \theta^*)^2\right] \tag{80}$$
$$= \frac{\omega\kappa^2}{2}\left[\mathrm{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2\right], \tag{81}$$

using the identity $\mathbb{E}_i[(\theta_i - \theta^*)^2] = \mathrm{Var}(\theta_i) + (\bar{\theta} - \theta^*)^2$. □

B.5 Proof of Theorem 3

Step 1: Dynastic criterion. Under the dynastic criterion, welfare is $W^{dyn} = \phi V_R + (1-\phi)V_M$ where each V_j is evaluated using household j's subjective beliefs. An announcement θ^A shifts beliefs to θ_j^{post}, which changes fertility choices. But households are optimizing given their (updated) beliefs, so any distortion in the announcement that moves beliefs away from households' priors makes them worse off according to their own criterion. The planner maximizes welfare by providing accurate information: $\theta_{dyn}^{A*} = \hat{\theta}$.

Step 2: Paternalistic criterion. Under the paternalistic criterion, welfare is evaluated under the true θ^*. The planner wants households to choose fertility as if they believed θ^*. The announcement that achieves this shifts household posteriors toward $\hat{\theta}$ (the planner's best estimate of θ^*). But the planner can do better by overshooting: if $\hat{\theta} > \bar{\theta}$, announcing slightly above $\hat{\theta}$ shifts beliefs further toward truth, improving welfare. The optimal shading reflects the tradeoff between moving mean beliefs toward truth and the quadratic loss from noise. Standard signal-extraction algebra gives the formula in the theorem.

Step 3: Total utilitarian criterion. Under total utilitarianism, the planner additionally values having more people. In a Romerian world, this pushes the optimal announcement toward θ_R to maximize fertility. The coefficient $\kappa_{tot} > \kappa_{pat}$ reflects this additional population motive. □

B.6 Ordering of Announcements (Auxiliary Result)

Corollary 3 (Ordering of Announcements)**.** *In a Romerian world where the planner's posterior mean $\hat{\theta}$ is close to θ_R:*

$$\theta_{dyn}^{A*} = \hat{\theta} \le \theta_{pat}^{A*} \le \theta_{tot}^{A*}. \tag{82}$$

The total utilitarian planner most aggressively promotes Romerianism; the dynastic planner is most conservative.

Proof. The first equality is part (i) of Theorem 3. The first inequality follows because $\kappa_{pat} > 0$ and, when $\hat{\theta}$ is close to θ_R, $\hat{\theta} - \bar{\theta} > 0$, so $\theta_{pat}^{A*} \geq \hat{\theta}$. The second inequality follows because $\kappa_{tot} > \kappa_{pat}$ and $\theta_R - \bar{\theta} \geq \hat{\theta} - \bar{\theta}$ when $\hat{\theta}$ is close to θ_R, giving $\theta_{tot}^{A*} \geq \theta_{pat}^{A*}$. □

B.7 State-Dependent Value of Announcement (Auxiliary Results)

Proposition 10 (Value of Announcement in the Trap)**.** *Suppose the economy is in the informational poverty trap region $\mathcal{T}(\delta, \eta)$ in a Romerian world. Let τ_ν^{trap} denote the planner's signal precision given the history of uninformative signals. Under paternalistic welfare, the expected value of truthful announcement is:*

$$\mathcal{V}^{trap}(\tau_\nu^{trap}) = P^{esc}(\tau_\nu^{trap}) \cdot \Delta W^{esc} - P^{deepen}(\tau_\nu^{trap}) \cdot \Delta W^{deepen}, \tag{83}$$

where P^{esc} is the probability that announcement initiates escape from the trap, ΔW^{esc} is the welfare gain from escape (large and positive), P^{deepen} is the probability that announcement deepens the trap, and ΔW^{deepen} is the welfare loss from deepening. There exists a threshold $\underline{\tau}^{trap}$ such that $\mathcal{V}^{trap} > 0$ if and only if $\tau_\nu^{trap} > \underline{\tau}^{trap}$.

Proposition 11 (Value of Announcement with Cycles)**.** *In a Malthusian world exhibiting cycles, under paternalistic welfare, the expected value of truthful announcement is:*

$$\mathcal{V}^{cycle}(\tau_\nu) = \Omega \cdot \big(Var(\theta_i) - \mathbb{E}[Var(\theta_i^{post})] \big) - (\textit{adjustment costs}), \tag{84}$$

where adjustment costs reflect the welfare loss from disrupting existing cyclical dynamics. For sufficiently precise signals, $\mathcal{V}^{cycle} > 0$, but the magnitude is smaller than $\mathcal{V}^{trap}$ when the trap gain ΔW^{esc} is large.

Proof of Propositions 10 and 11. These follow from combining the welfare decomposition with the state-dependent value of coordination. In the trap, the gain from escape ΔW^{esc} is large (avoiding extended stagnation), so even a moderately noisy announcement is valuable if it has positive probability of triggering escape. In cycles, the gain is smaller (dampening oscillations), and adjustment costs from disrupting cyclical dynamics partially offset benefits. The threshold precision $\underline{\tau}^{trap}$ is determined by equating expected gain and expected loss. Formal derivations follow the same quadratic approximation techniques used in the proof of the welfare-loss lemma above. □